

# Algebraic geometry 2026

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# Chapter 1

## Mynotes from 2026 class

Some references used here:

- Mumford's red book [3]
- Humphreys Linear algebraic groups [2]

### 1.1 Lecture 1 and 2

First some definitions and lemmas worth to remember.

- (Integral extension). Let  $R$  be a ring and  $S \subset R$  a subring. We say that  $R$  is integrally dependent over  $S$  if every  $r \in R$  is integral over  $S$ , that is, there exist a monic polynomial  $f \in S[X]$  such that  $f(r) = 0$ .

**Lemma 1.1.1.** Let  $S \subset R$  be rings. A finite subset of elements  $b_1, \dots, b_m \in R$  are all integral over  $S$  if and only if  $S[b_1, \dots, b_m]$  is finitely generated (as a  $S$ -module) over  $S$ .

Proof: ( $\Rightarrow$ ) Let  $b \in R$  be integral over  $S$ , i.e., there exist

$$f = X^d + a_1X^{d-1} + \dots + a_d \in S[X] \quad (1.1)$$

such that  $f(b) = 0$ . Then, using long division, we see that  $S[b] = S + \dots + Sb^{d-1}$  which is finitely generated. Now, if  $b_1, \dots, b_m$  are all integral over  $S$ , we can argue by induction that  $S[b_1, \dots, b_{m-1}]$  is finitely generated over  $S$ . However, since  $b_m$  is integral over  $S[b_1, \dots, b_{m-1}]$ ,  $S[b_1, \dots, b_m]$  is finitely generated over  $[b_1, \dots, b_{m-1}]$  and it must be finitely generated over  $S$  aswell.

( $\Leftarrow$ ) Suppose that  $S[b_1, \dots, b_m] = S\beta_1 + \dots + S\beta_n$  for some  $\beta_i \in R$ . Then for any  $b \in R$  which fixes  $S[b_1, \dots, b_m]$ , we have  $b\beta_i = \sum_j a_{ij}\beta_j$  for some  $(a_{ij}) \in \text{Mat}_n(S)$ . Therefore

$$0 = (bI - (a_{ij}))^*(bI - (a_{ij}))v = \det(bI - (a_{ij}))v \quad (1.2)$$

where  $v = (\beta_1, \dots, \beta_n)^T$ . Therefore  $\det(bI - (a_{ij}))\beta_i = 0, \forall i$  and since  $1 \in S\beta_1 + \dots + S\beta_n$ , we see that  $\det(bI - (a_{ij})) = 0$ . Since  $\det(X - (a_{ij})) \in S[X]$  is monic, we are done.  $\square$

**Corollary 1.1.2.** (Transitive property of integral dependence). Given rings  $R \supset S \supset T$ , if  $R$  is integral over  $S$  and  $S$  is integral over  $T$ , then  $R$  is integral over  $T$ .

**Question:** Do we need Noetherian hypothesis using this statement?

Proof: Let  $r \in R$ , then by hypothesis there exist  $f = X^d + s_1X^{d-1} + \dots \in S[X]$  such that  $f(r) = 0$ . Define  $A = T[s_1, \dots, s_d]$ , then  $A[r]$  is a finitely generated  $A$ -module because  $S \subset A$  and lemma (1.1.1). Now, since  $A$  is a finitely generated  $T$ -module by lemma (1.1.1), we see that  $A[r]$  is a finitely generated  $T$ -module. Now, **assuming  $T$  is Noetherian** we see that  $T[r]$  is finitely generated as a  $T$ -module.  $\square$

**Remark 1.1.3.** We can certainly say (using the previous proof) that if  $r \in R$ , then  $S[r]$  is finitely generated as a  $T$ -module.

**Lemma 1.1.4.** Let  $K \supset k$  ( $k$  a perfect field) be a finitely generated extension field with transcendence degree  $\partial_k(K) = d$ , then there exist generators  $z_1, \dots, z_d, z_{d+1}$  such that

- (1)  $K = k(z_1, \dots, z_{d+1})$
- (2)  $\{z_1, \dots, z_d\}$  are algebraically independent
- (3)  $z_{d+1}$  is separable over  $k(z_1, \dots, z_d)$

**Lemma 1.1.5.** (Noether's normalization lemma - Mumford's red book). Let  $R$  be a finitely generated algebra over  $k$ . Assume also that  $R$  is an integral domain. If  $R$  has transcendence degree  $n$  over  $k$  (which means that  $\partial_k \text{Frac}(R) = n$ ), then there exist  $x_1, \dots, x_n \in R$  algebraically independent and such that  $R$  is integral over  $k[x_1, \dots, x_n]$ .

**TODO:** Study prove this version of the lemma using the statement made in class.

**Lemma 1.1.6.** (Noether's normalization lemma - Statement made in class). Let  $R$  be a finitely generated  $k$ -algebra. Then there exist  $x_1, \dots, x_n \in R$  algebraically independent over  $k$  such that  $R$  is integral over  $k[x_1, \dots, x_n]$ .

Proof (Nagata): Since  $R$  is finitely generated as a  $k$ -algebra, there exist generators  $y_1, \dots, y_m$  such that  $R = k[y_1, \dots, y_m]$ . Now suppose that  $y_1, \dots, y_m$  are algebraically independent, then we can take  $x_i = y_i$ . If not, then there exist  $p \in k[Y_1, \dots, Y_m] \setminus 0$  such that  $p(y_1, \dots, y_m) = 0$ . Write

$$p(Y_1, \dots, Y_m) = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^m} a_\alpha Y_1^{\alpha_1} \dots Y_m^{\alpha_m} \quad (1.3)$$

and define a new polynomial

$$\begin{aligned} q(Y_1, \dots, Y_m) &= p(Y_1, Y_2 + Y_1^{r_2}, \dots, Y_m + Y_1^{r_m}) \\ &= \sum_{\alpha} a_{\alpha} (Y_1^{\alpha_1 + r_2 \alpha_2 + \dots + r_m \alpha_m} + \dots) \end{aligned} \quad (1.4)$$

for some choice of positive integers  $r_2, \dots, r_m$ . Take a positive number  $b > \alpha_i, \forall \alpha_i$  such that  $a_{\alpha} \neq 0$ . If we take  $r_i = b^i$ , then  $\alpha_1 + \alpha_2 b + \dots + \alpha_m b^{m-1}$  are all different numbers in base  $b$ . So, from different powers of the form  $\alpha_1 + r_2 \alpha_2 + \dots + r_m \alpha_m$  we have a maximal one  $d = \max\{\alpha_1 + \dots + r_m \alpha_m : a_{\alpha} \neq 0\}$  and

$$q(Y_1, \dots, Y_m) = aY_1^d + \text{smaller degree terms} \in k[Y_2, \dots, Y_m][Y_1], \quad a \neq 0. \quad (1.5)$$

Now, if we let  $z_i = y_i - y_1^{r_i} \in R, i = 2, \dots, m$ , we see that  $q(y_1, z_2, \dots, z_m) = p(y_1, \dots, y_m) = 0$  and  $y_1$  is integral over  $k[z_2, \dots, z_m]$ . By induction on number of generators, there exist  $x_1, \dots, x_n$  algebraically independent such that  $k[z_2, \dots, z_m]$  is integral over  $k[x_1, \dots, x_n]$ . So, by lemma 1.1.1 and its corollary (1.1.1), we see that  $k[y_1, z_2, \dots, z_m]$  is integral over  $k[x_1, \dots, x_m]$ . Now, by definition, it's clear that  $k[y_1, z_2, \dots, z_m]$  is a finitely generated  $R$ -module. So by the corollary (1.1.2) again, we see that  $R$  is integral over  $k[x_1, \dots, x_m]$ .  $\square$

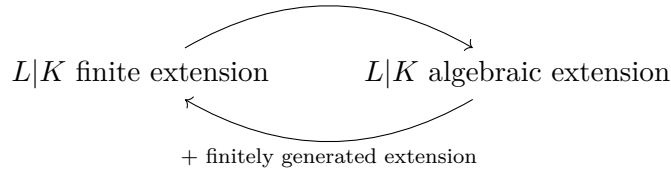
**Lemma 1.1.7.** Let  $S \subset R$  be a ring extension. If  $R$  is a field, which integral over  $S$ , then  $S$  is a field.

Proof: Let  $s \in S \setminus 0$  and  $r \in R$  be its inverse. Since  $R$  is integral over  $S$ , there exist  $f = X^d + a_1 X^{d-1} + \dots \in S[X]$  be such that  $f(r) = 0$ . Therefore

$$0 = s^{d-1} f(r) = r + a_1 + sa_2 + \dots + s^{d-1} a_0 \Rightarrow r \in S. \quad (1.6)$$

$\square$

**Remark 1.1.8.** Let  $L|K$  be a field extension. Then



**Theorem 1.1.9.** (Zariski). Let  $k$  be a field and  $\mathfrak{m} \subset R = k[X_1, \dots, X_m]$  be a maximal ideal. Then  $L = R/\mathfrak{m}$  is a finite extension field of  $k$ .

Proof: By Noether normalization lemma (1.1.6), there exist  $a_1, \dots, a_n \in L$  algebraically independent such that  $L$  is integral over  $k[a_1, \dots, a_n]$ . By lemma (1.1.7), we conclude

that  $k[a_1, \dots, a_n]$  is a field. Suppose that  $n > 0$ , so  $a_1$  has an inverse. This means that  $a_1 g(a_1, \dots, a_n) = 1$  for some  $g \in k[X_1, \dots, X_n]$ , which contradicts the fact that  $a_1, \dots, a_n$  are algebraically independent. Then we conclude that  $L$  is an integral (and then algebraic) over  $k$ . However, since  $L$  is a finitely generated extension ( $L = k(\overline{X}_1, \dots, \overline{X}_m)$ ) over  $k$  we conclude that  $L$  is in fact a finite field extension over  $k$ .  $\square$

**Theorem 1.1.10.** Suppose that  $k$  is an algebraically closed field. Let  $I \subset R = k[X_1, \dots, X_m]$  be an ideal, then  $V(I) = \emptyset \iff I = k[X_1, \dots, X_m]$ .

Proof: If  $I \neq R$ , then there exist a maximal ideal  $\mathfrak{m} \supset I$ . So  $V(I) \supset V(\mathfrak{m})$  and it's sufficient to prove that  $V(\mathfrak{m}) \neq \emptyset$ . By theorem (1.1.9), we have that  $L = R/\mathfrak{m}$  is a finite  $k$ -extension, so we must have  $L = k$  because  $k$  is algebraically closed. So there exist  $a_i \in k, i = 1, \dots, m$  such that  $X_i - a_i \in \mathfrak{m}$  and  $(X_1 - a_1, \dots, X_m - a_m) \subset \mathfrak{m}$ . However  $\varphi : R \rightarrow k : F \mapsto F(a_1, \dots, a_m)$  has kernel  $\text{Ker}(\varphi) = (X_1 - a_1, \dots, X_m - a_m)$  and since  $k$  is a field,  $R/\text{Ker}(\varphi)$  must be a field and  $\text{Ker}(\varphi)$  is a maximal ideal. Therefore we conclude that  $\mathfrak{m} = (X_1 - a_1, \dots, X_m - a_m)$ . Finally  $V(X_1 - a_1, \dots, X_m - a_m) = \{a_1, \dots, a_m\} \neq \emptyset$ .  $\square$

**Remark 1.1.11.** The hypothesis  $k = \overline{k}$  in the previous theorem is crucial. For example if  $k = \mathbb{R}$ , we have  $V(x^2 + 1) = \emptyset$ .

**Corollary 1.1.12.** From the proof of the previous theorem, we see that there exist a bijection

$$\begin{aligned} k^n &\longleftrightarrow \left\{ \begin{array}{c} \text{maximal ideals} \\ \mathfrak{m} \subset R \end{array} \right\} \\ (a_1, \dots, a_n) &\longmapsto (X_1 - a_1, \dots, X_n - a_n) \end{aligned} \quad (1.7)$$

**Theorem 1.1.13.** (Hilbert Nullstellensatz). Let  $A \subset k[X_1, \dots, X_n]$  be an ideal, then

$$IV(A) = \sqrt{A}. \quad (1.8)$$

Proof: The inclusion  $\sqrt{A} \subset IV(A)$  is easy. Now, assume  $g \in IV(A)$ , where  $A = (f_1, \dots, f_m)$ . This means that we have the hypothesis:

$$f_i(a_1, \dots, a_n) = 0, \forall i = 1, \dots, m \Rightarrow g(a_1, \dots, a_n) = 0 \quad (*) \quad (1.9)$$

Now, define the ideal

$$I = Ak[X_1, \dots, X_{n+1}] + (1 - g(X_1, \dots, X_n)X_{n+1}). \quad (1.10)$$

If  $I$  is a proper ideal, then by corollary (1.1.12), we have  $I \subset (X_1 - a_1, \dots, X_{n+1} - a_{n+1})$ , which is the kernel of the evaluation map  $k[X_1, \dots, X_{n+1}] \rightarrow k : f \mapsto f(a_1, \dots, a_{n+1})$ . This means that

$$\begin{cases} f_i(a_1, \dots, a_n) = 0, \forall i = 1, \dots, m \\ g(a_1, \dots, a_n)a_{n+1} = 1 \end{cases} \quad (1.11)$$

which contradicts our initial hypothesis (\*). Therefore,  $I$  is not a proper ideal, so  $1 \in I$  and there exists  $h_1, \dots, h_{m+1} \in k[X_1, \dots, X_{n+1}]$  such that

$$1 = \sum_{j=1}^m h_j(X_1, \dots, X_{n+1}) f_j(X_1, \dots, X_n) + h_{m+1}(X_1, \dots, X_{n+1})(1 - g(X_1, \dots, X_n)X_{n+1}). \quad (1.12)$$

Now, by projecting this equation onto  $k[X_1, \dots, X_{n+1}]/(1 - gX_{n+1})$ , we get

$$g^\ell(X_1, \dots, X_n) = \sum_{j=1}^m h_j(X_1, \dots, X_{n+1}) g^\ell \times f_j(X_1, \dots, X_n) \quad (1.13)$$

modulo  $(1 - gX_{n+1})$ , where  $H_j = h_j(X_1, \dots, X_{n+1}) g^\ell \in k[X_1, \dots, X_n]$ ,  $\forall j$  if  $\ell \gg 0$ . Since  $\{\bar{X}_1, \dots, \bar{X}_n\}$  is algebraically free in  $k[X_1, \dots, X_{n+1}]/(1 - gX_{n+1})$ , we conclude that, in fact, we have equality

$$g^\ell(X_1, \dots, X_n) = \sum_{j=1}^m H_j(X_1, \dots, X_n) \times f_j(X_1, \dots, X_n) \quad (1.14)$$

in  $k[X_1, \dots, X_n]$ . □

**Remark 1.1.14.** We could have used the homomorphism

$$k[X_1, \dots, X_{n+1}] \rightarrow k(X_1, \dots, X_n) : X_{n+1} \mapsto g, \quad X_i \mapsto X_i, \quad i \neq n+1 \quad (1.15)$$

at the end of the previous proof.

## 1.2 Lecture 3

**Definition 1.2.1.** We define an affine algebraic set  $V \subset k^n$  to be a set of the form  $V = V(S)$  for some  $S \subset k[\underline{X}]$ .

**Notation 1.2.2.** Let  $V$  be an affine algebraic set. We write

$$D_V(g) = \{p \in V : g(p) \neq 0\} = V \setminus V(g). \quad (1.16)$$

These open sets are called “principal open subsets”.

**Lemma 1.2.3.** Any open subset  $U \subset V$  of an affine algebraic set is quasicompact.

Proof: Consider a covering  $U = \bigcup_{\lambda \in \Lambda} U_\lambda$ . Observe that  $V \setminus U = V(g_1, \dots, g_m)$ , so we have the equality  $U = \bigcup_{i=1}^m D_V(g_i)$ . In particular, we see that principal open subsets form a basis for the Zariski topology on  $V$ . Therefore, it's sufficient to prove that a principal

open subset is quasicompact. So we assume  $U = D_V(g)$ . Furthermore, we have  $U_\lambda = \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i})$  and

$$U = \bigcup_{\lambda \in \Lambda} \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i}). \quad (1.17)$$

So, without loss of generalization, we can assume  $U = \bigcup_{\lambda \in \Lambda} D_V(g_\lambda)$  from beginning. Now

$$\begin{aligned} D_V(g) &= \bigcup_{\lambda \in \Lambda} D_V(g_\lambda) \iff \\ V \cap V(g) &= V \cap V(\langle g_\lambda : \lambda \in \Lambda \rangle) \iff \\ V \cap V(g) &= V \cap V(\langle g_\lambda : \lambda \in I \rangle) \quad (I \subset \Lambda \text{ finite}). \end{aligned} \quad (1.18)$$

In the last equation we used Hilbert basis theorem: The sequence

$$\langle g_{\lambda_1} \rangle \subset \langle g_{\lambda_1}, g_{\lambda_2} \rangle \subset \cdots \quad (1.19)$$

must terminate at some point, because  $k[\underline{X}]$  is Noetherian. Then we conclude that  $D_V(g) = \bigcup_{\lambda \in I} D_V(g_\lambda)$ .  $\square$

**Lemma 1.2.4.** Suppose that  $k$  is an infinity field, then open sets  $U \subset k^n$  are dense in the Zariski topology.

Proof: Since basic open sets form a basis for the Zariski topology on  $k^n$ , it's sufficient to prove that  $D(f)$  is dense in  $k^n$ . Let  $\overline{D(f)} = V(g_1, \dots, g_m)$ ,  $m \geq 1$  and  $x \in k^n$ , then we have for all  $i = 1, \dots, m$ :

$$f(x) \neq 0 \Rightarrow g_i(x) = 0 \Rightarrow fg_i(x) = 0, \forall x \in k^n. \quad (1.20)$$

Therefore, since  $k$  is infinity, we have  $fg_i = 0 \in k[\underline{X}]$ ,  $\forall i$  and  $g_i = 0$ ,  $\forall i$ . Then we see that  $\overline{D(f)} = k^n$ .  $\square$

**Definition 1.2.5.** (Regular functions). Let  $V \subset k^n$  be an affine algebraic set and  $U \subset V$  be an open set. A map  $\varphi : U \rightarrow k$  is said to be *regular* if for any  $p \in U$ , there exist  $p \in U' \subset U$ , polynomials  $f, g \in k[\underline{X}]$  such that  $g(q) \neq 0$ ,  $\forall q \in U'$  and  $\varphi|_{U'} = f/g|_{U'}$ .

**Notation 1.2.6.** Let  $V$  be an affine algebraic set and let  $U \subset V$  be an open set, we denote  $\mathcal{O}_V(U) = \{U \xrightarrow{\text{regular}} k\}$ .



**Lemma 1.2.7.** Let  $V \subset k^n$  be an affine algebraic set and take  $g \in k[X_1, \dots, X_n] \setminus I(Y)$ , so  $D_V(g) \neq \emptyset$ . Then the set of regular functions  $\varphi : D_V(g) \rightarrow k$  is of the form  $f/g^m|_{D_V(g)}$  for some  $f \in k[X_1, \dots, X_n]$  and  $m > 0$ . Then we have a surjection

$$\tilde{\psi} : k[V]_{\tilde{g}} \rightarrow \mathcal{O}_V(D_V(g)) \quad (1.21)$$

where  $\psi : k[V] \rightarrow \mathcal{O}_V(D_V(g))$  is the map that takes a polynomial and restricts it to  $D_V(g)$  and  $\tilde{\psi}$  is the corresponding map induced by the universal property of localization. Furthermore, if  $V$  is irreducible, this map is an isomorphism.

Proof: First we have  $D_V(g) = \bigcup_{i=1}^{\ell} D_V(g_i)$  and since  $\varphi$  is regular, there exists  $f_i, h_i \in k[\underline{X}]$ ,  $i = 1, \dots, \ell$  such that

$$q \in D_V(g_i) \Rightarrow h_i(q) \neq 0 \text{ (so } q \in D_V(h_i)), \varphi|_{D_V(g_i)} = \frac{f_i}{h_i} \Big|_{D_V(g_i)}. \quad (1.22)$$

Now, we have two fundamental relations:

$$\begin{cases} D_V(g_i) \subset D_V(h_i), \forall i = 1, \dots, \ell \\ D_V(g) = \bigcup_{i=1}^{\ell} D_V(g_i) \end{cases} \quad (1.23)$$

Using  $V = V(F_1, \dots, F_m)$  and Hilbert Nullstellensatz (1.1.13), we get

$$\begin{cases} \sqrt{(F_1, \dots, F_m, g_i)} \subset \sqrt{(F_1, \dots, F_m, h_i)}, \forall i = 1, \dots, \ell & (*) \\ \sqrt{(F_1, \dots, F_m, g)} = \sqrt{(F_1, \dots, F_m, g_1, \dots, g_{\ell})} & (**) \end{cases} \quad (1.24)$$

From (\*), we deduce that  $g_i^{n_i} = \sum_j A_j F_j + B_i h_i$  and restricting to  $V$  we get  $g_i^{n_i} = B_i h_i$ . Now, since  $\varphi|_{D_V(g_i)} = \frac{B_i f_i}{g_i^{n_i}}$  and  $D_V(g_i) = D_V(g_i^{n_i})$ , we can **change notation**  $B_i f_i \rightarrow f_i$  and  $g_i^{n_i} \rightarrow g_i$ .

Now we do a trick: since  $g_j f_i - f_j g_i = 0$  on  $D_V(g_i) \cap D_V(g_j) = D_V(g_i g_j)$ , we get

$$g_i g_j (g_j f_i - f_j g_i) = 0 \text{ on } V. \quad (1.25)$$

Furthermore, we still have  $\varphi|_{D_V(g_i)} = g_i f_i / g_i^2$ . So we **change notation** again:  $g_i^2 \rightarrow g_i$  and  $g_i f_i \rightarrow f_i$ , so we have (on  $V$ )

$$g_i f_j - g_j f_i = 0, \forall i, j = 1, \dots, \ell \quad (***). \quad (1.26)$$

Now, by equation (\*\*), there exist  $m > 0$  such that

$$g^m = \sum_j A_j F_j + \sum_k B_k g_k \quad (1.27)$$

and we get on  $V$ :

$$g^m = \sum_k B_k g_k. \quad (1.28)$$

Now, by using equation  $(***)$ , we have (on  $V$ )

$$f_i g^m = \sum_k B_k f_i g_k = \sum_k B_k f_k g_i = \left( \sum_k B_k f_k \right) g_i. \quad (1.29)$$

So if  $q \in D_V(g_i)$ , we have  $\varphi(q) = \frac{f_i}{g_i}(q) = \frac{f}{g^m}(q)$ , where we set  $f = \sum_k B_k f_k$ .

This proves that  $\hat{\psi}$  is surjective. To prove injectivity, we observe that  $f/g^m|_{D_V(g)} = 0 \iff f|_{D_V(g)} = 0 \iff fg|_V = 0$ . So if  $V$  is irreducible, then  $fg \in I(Y) \Rightarrow f \in I(Y)$  because  $I(Y)$  is prime, so  $\bar{f} = 0$ .  $\square$

**Corollary 1.2.8.** Since  $V = D_V(1)$ , we have

$$\mathcal{O}_V(V) \cong k[V]_{(1)} \cong k[V]. \quad (1.30)$$

**Definition 1.2.9.** A morphism between two affine algebraic sets  $V_1 \subset k^n$  and  $V_2 \subset k^m$  is a function  $\varphi : V_1 \rightarrow V_2$  such that for any open set  $U \subset V_2$ , we have

$$\varphi_U^\# : \mathcal{F}(U, k) \xrightarrow{-\circ\varphi} \mathcal{F}(\varphi^{-1}(U), k), \quad \varphi_U^\#(\mathcal{O}_{V_1}(U)) \subset \mathcal{O}_{V_2}(\varphi^{-1}(U)). \quad (1.31)$$

Using more fancy words, the previous definition says that a continuous map  $\varphi : V_1 \rightarrow V_2$  is a morphism if the composition map  $\varphi_U^\# : \mathcal{F}(U, k) \xrightarrow{-\circ\varphi} \mathcal{F}(\varphi^{-1}(U), k)$  induces a sheaf morphism

$$\varphi^\#(U) : \mathcal{O}_{V_2} \rightarrow f_* \mathcal{O}_{V_1}. \quad (1.32)$$

We use the notation  $f_* \mathcal{O}_{V_1}$  for direct image sheaf, it's defined by  $f_* \mathcal{O}_{V_1}(U) = \mathcal{O}_{V_1}(\varphi^{-1}(U))$ .

### 1.3 Lecture 4

**Definition 1.3.1.** A function space is a topological space  $X$  with a sheaf  $\mathcal{O}_X$  such that for any open set  $U \subset X$ , we have

- (1)  $\mathcal{O}_X(U) \overset{\text{subring}}{\subset} \{U \xrightarrow{\text{continuous}} k\}$
- (2) Let  $f \in \mathcal{O}_X(U)$ , then  $f$  has an inverse in  $\mathcal{O}_X(U)$  if and only if  $f(q) \neq 0, \forall q \in U$ .

**Lemma 1.3.2.** An affine algebraic set  $V \subset k^n$  equipped if the sheaf of regular functions is a function space.

Proof: Let  $U \subset X$  be an open set and  $f \in \mathcal{O}_X(U)$ . Then, by definition, there exist an open cover  $U = \bigcup_\lambda U_\lambda$  such that  $f|_{U_\lambda} = f_\lambda/g_\lambda$ , where  $g_\lambda(q) \neq 0 \forall q \in U_\lambda$ . To prove that  $f$  is continuous, it's sufficient to prove that  $f^{-1}(a) \subset V$  is closed for all  $a \in k$ . So we prove that  $(f|_{U_\lambda})^{-1}(a)$  is closed in  $V$  for all  $\lambda$ , which can be verified by the equality

$$V \cap V(f_\lambda - ag_\lambda) = (f|_{U_\lambda})^{-1}(a). \quad (1.33)$$

The second item is clear.  $\square$

**Definition 1.3.3.** Let  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$  be two function spaces. We define a morphism  $X \rightarrow Y$  to be a continuous map  $f : X \rightarrow Y$  which induces a sheaf map  $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$  in the sense that if  $f^\#(U)(g) = g \circ f$  for  $g \in \mathcal{O}_Y(U)$ , then  $f^\#(U)(\mathcal{O}_Y(U)) \subset \mathcal{O}_X(f^{-1}(U))$ .

**Remark 1.3.4.** Being a morphism of function spaces is a local property. In fact, let  $f : X \rightarrow Y$  be a morphism of function spaces and suppose that  $Y = \bigcup_\lambda V_\lambda$  and for each  $\lambda$ :

$$f|_{f^{-1}(V_\lambda)} : f^{-1}(V_\lambda) \rightarrow V_\lambda \quad (1.34)$$

is a morphism. Then  $f$  is a morphism.

**Definition 1.3.5.** An affine algebraic variety is a function space  $(X, \mathcal{O}_X)$  isomorphic to an affine algebraic set with Zariski topology and sheaf of regular functions.

**Definition 1.3.6.** Let  $(X, \mathcal{O}_X)$  be a function space, and  $U \subset X$  be an open set with induced topology and  $\mathcal{O}_U = \mathcal{O}_X|_U$ . Then  $(U, \mathcal{O}_U)$  is a function space called induced open subspace.

**Definition 1.3.7.** An abstract algebraic variety is a function space  $(X, \mathcal{O}_X)$  such that there exist an open covering  $X = \bigcup_\lambda U_\lambda$  with the property that for each  $\lambda$ , the induced open subspace  $(U_\lambda, \mathcal{O}_{U_\lambda})$  is an affine algebraic variety.

**Remark 1.3.8.** In many texts we have the additional hypothesis of irreducibility in the definition of algebraic varieties.

**Definition 1.3.9.** An affine algebraic variety is said to be quasi-affine if it's isomorphic, as a function space, to an induced open subspace of an affine algebraic variety.

**Lemma 1.3.10.** Let  $(X, \mathcal{O}_X)$  be a function space and  $Y \subset k^n$  be a quasi affine algebraic set

**Lemma 1.3.11.** Let  $(X, \mathcal{O}_X)$  be a function space and  $Y \subset k^n$  be a quasi-affine algebraic set. Then  $f : X \rightarrow Y$  is a morphism if and only  $f = (g_1, \dots, g_n)$  with  $g_i \in \mathcal{O}_X(X)$ . Here, we think  $Y$  as an induced open subspace of an affine algebraic set.

Proof:  $(\Rightarrow)$  The coordinate functions  $x_i|_Y : Y \rightarrow k$  are regular functions, by definition. Now  $g_i = x_i \circ f = f^\#(Y)(x_i) \in \mathcal{O}_X(f^{-1}(Y)) = \mathcal{O}_X(X)$ , because  $x_i \in \mathcal{O}_Y(Y) = \{Y \xrightarrow{\text{regular}} k\}$ .  
 $(\Leftarrow)$  A closed set of  $Y$  is of the form  $V(F) \cap Y$  (would not be of the form  $V(F_1, \dots, F_m)$  in general ?). However  $f^{-1}(V(F) \cap Y) = \{p \in X : F(g_1, \dots, g_n)(p) = 0\}$  which is closed in  $X$  provided we show that  $F(g_1, \dots, g_n)$  is continuous. However, since  $\mathcal{O}_X(X)$  is a ring, we have  $F(g_1, \dots, g_n) \in \mathcal{O}_X(X)$ . Furthermore, if  $U \subset Y$  is an open set and  $\varphi : U \rightarrow k$  is regular, we need to show that  $f^\#(U)(\varphi) \in \mathcal{O}_X(f^{-1}(U))$ . Let  $U' \subset U$  be such that  $\varphi|_{U'} = F/G$ ,  $G(q) \neq 0 \forall q \in U'$ . Then we have

$$f^\#(U')(\varphi) = \varphi \circ f|_{f^{-1}(U')} = \frac{F(g_1, \dots, g_n)}{G(g_1, \dots, g_n)} \Big|_{f^{-1}(U')}. \quad (1.35)$$

Furthermore

$$G \circ f(q) = G(g_1, \dots, g_n)(q) \neq 0, \forall q \in f^{-1}(U'). \quad (1.36)$$

Since  $G(g_1, \dots, g_n)|_{f^{-1}(U')} \in \mathcal{O}_X(f^{-1}(U'))$ , the previous equation shows that the inverse  $1/G(g_1, \dots, g_n)$  also is an element of  $\mathcal{O}_X(f^{-1}(U'))$ . Now, let  $U = \bigcup_\lambda U_\lambda$ ,  $\varphi|_{U_\lambda} = F_\lambda/G_\lambda$ , then we see that

$$f^\#(U_\lambda)(\varphi|_{U_\lambda}) \subset \mathcal{O}_X(f^{-1}(U_\lambda)). \quad (1.37)$$

The glueing axiom shows that  $f^\#(U)(\varphi) \in \mathcal{O}_X(f^{-1}(U))$ .  $\square$

**Lemma 1.3.12.** Let  $V \subset k^n$  be an affine algebraic set and  $D_V(F) \subset V$  an affine algebraic set. Then the induced open subspace  $(D_V(F), \mathcal{O}_{D_V(F)})$  is a quasi-affine algebraic variety, by definition. However, we have more, it's also an affine algebraic variety.

Proof: Let  $V = V(G_1, \dots, G_m)$  and define the map

$$\varphi : D_V(F) \rightarrow V(x_0 F - 1, G_1, \dots, G_m) \subset k^{n+1} : (a_1, \dots, a_n) \mapsto \left( \frac{1}{F(a_1, \dots, a_n)}, a_1, \dots, a_n \right) \quad (1.38)$$

By the lemma (1.3.11), we see that  $\varphi$  is a morphism. Furthermore, it has a set theoretical inverse given by the projection

$$\psi : V(x_0 F - 1, G_1, \dots, G_m) \rightarrow D_V(F) : (a_0, \dots, a_n) \mapsto (a_1, \dots, a_n). \quad (1.39)$$

By the same lemma, this map is also a morphism. Since  $f\psi = \mathbb{1}_{D_V(x_0 F - 1, \dots)}$  and  $\psi f = \mathbb{1}_{D_V(F)}$ , we see that  $\varphi$  is an isomorphism (the induced sheaf morphism would be an isomorphism by construction).  $\square$

**Remark 1.3.13.** Let  $(X, \mathcal{O}_X)$  be an affine algebraic variety and  $\varphi : (V, \mathcal{O}_V) \rightarrow (X, \mathcal{O}_X)$  an isomorphism, where  $V \subset k^n$  is an algebraic set. Then this isomorphism, induces another isomorphism

$$(D_V(F), \mathcal{O}_{D_V(F)}) \rightarrow (X_f, \mathcal{O}_{X_f}) \quad (1.40)$$

where  $F = f \circ \varphi$  and  $X_f = \varphi(D_V(F)) = \{p \in X : f(p) \neq 0\}$ .

**Corollary 1.3.14.** A quasi-affine algebraic variety is an abstract algebraic variety.

Proof: A quasi-affine algebraic variety has a topological basis consisting of principal open subsets, which are affine algebraic varieties.

## 1.4 Lecture 5 and 6

**Theorem 1.4.1.** Suppose that we have the data:

- $(X_\lambda)_{\lambda \in \Lambda}$  function spaces over  $k$
- $\forall \lambda, \mu \in \Lambda$  let  $U_{\lambda\mu} \subset X_\lambda$
- $\forall \lambda, \mu \in \Lambda$  an isomorphism

$$\varphi_{\lambda\mu} : U_{\lambda\mu} \rightarrow U_{\mu\lambda} \quad (1.41)$$

Furthermore, assume that

- $U_{\lambda\lambda} = X_\lambda, \varphi_{\lambda\lambda} = 1_{X_\lambda}$
- $\forall \lambda, \mu, \nu \in \Lambda$ , we have

$$\begin{aligned} \varphi_{\lambda\mu}(U_{\lambda\mu} \cap U_{\lambda\nu}) &\subset U_{\mu\lambda} \cap U_{\mu\nu} \\ \varphi_{\mu\nu}\varphi_{\lambda\mu}|_{U_{\lambda\mu} \cap U_{\lambda\nu}} &= \varphi_{\lambda\nu}|_{U_{\lambda\mu} \cap U_{\lambda\nu}} \end{aligned} \quad (1.42)$$

Then,  $X = \coprod_{\lambda \in \Lambda} X_\lambda / \sim$ , where we put  $(\lambda, x) \sim (\mu, x')$  if and only if  $\varphi_{\lambda\mu}(x) = x'$ . One can verify that this defines an equivalence relation. Furthermore,  $X$  has a topology defined as the small topology such that all the maps  $i_\lambda : X_\lambda \rightarrow X : x \mapsto [(\lambda, x)]$  are continuous. We define a sheaf

$$\mathcal{O}_X(U) = \{f : U \rightarrow k : f i_\lambda \in \mathcal{O}_{X_\lambda}(i_\lambda^{-1}(U)), \forall \lambda \in \Lambda\}. \quad (1.43)$$

With this sheaf,  $(X, \mathcal{O}_X)$  is a function space satisfying the following universal property: for any function space  $Y$  and morphisms  $j_\lambda : X_\lambda \rightarrow Y$  such that:

$$j_\mu \varphi_{\lambda\mu} = j_\lambda|_{U_{\lambda\mu}}, \quad (1.44)$$

there exist a unique morphism  $f : X \rightarrow Y$  such that  $f i_\lambda = j_\lambda$  (morphisms here, means morphisms of function spaces!). Furthermore, since  $i_\mu^{-1} i_\lambda(X_\lambda) = U_{\mu\lambda} \subset X_\mu, \forall \mu$ , then  $i_\lambda(U_\lambda)$  is open and  $i_\lambda$  are isomorphisms onto its images. So, if each  $X_\lambda$  is an affine algebraic variety, then  $X$  is an abstract variety.

Proof: Tedious. □

**Definition 1.4.2.** An abstract variety  $X$  is separable if for any pair of affine open sets  $U, V \subset X$ , we have

- (1)  $U \cap V$  is affine
- (2) The induced map

$$\mathcal{O}(U) \otimes \mathcal{O}(V) \rightarrow \mathcal{O}(U \cap V) : f \otimes g \mapsto f|_{U \cap V} \cdot g|_{U \cap V} \quad (1.45)$$

is surjective.

**Example 1.4.3.** Let  $X_0 = X_1 = k^1$  and  $U_{01} = U_{10} = k^1 - \{0\}$ . Define a map  $\varphi_{01} : U_{01} \rightarrow U_{10} : x \mapsto 1/x$ . We claim that  $X = \mathbb{P}^1$  in the variety obtained by glueing  $X_0$  and  $X_1$  through  $\varphi_{01}$ . In fact, for  $i = 0, 1$ , let  $g_i : X_i \rightarrow Y$  be two morphisms such that

$$g_1 \varphi_{01} = g_0|_{U_{01}}. \quad (1.46)$$

That is,  $g_1(1/x) = g_0(x)$ . Then the following map is well defined:

$$\begin{aligned} g : X &\rightarrow Y \\ [1 : y] &\mapsto g_0(y) \\ [x : 1] &\mapsto g_1(x) \end{aligned} \quad (1.47)$$

Moreover, by construction we have  $g|_{U_i} = g_i \varphi_i$ , where  $\varphi_i : U_i \rightarrow k^1$  (here,  $U_i = \{[x_0, x_1] : x_i \neq 0\}$ ) are the standard charts of the projective plane  $X$ . So we see that  $g$  is locally a morphism, i.e., it's a morphism.

Is  $X$  separable? Consider the following diagram:

$$\begin{array}{ccccc} \mathcal{O}_X(U_0) \otimes \mathcal{O}_X(U_1) & \xrightarrow{\quad} & \mathcal{O}_X(U_0 \cap U_1) & & \\ \uparrow & & \downarrow & & \\ \mathcal{O}_{k^1}(k^1) \otimes \mathcal{O}_{k^1}(k^1) & & \mathcal{O}_{k^1}(k^1 - \{0\}) & & \\ & \nearrow \varphi_0^\#(s) \otimes \varphi_1^\#(t) & & \xrightarrow{\quad} & \varphi_0^\#(s)|_{U_0 \cap U_1} \varphi_1^\#(t)|_{U_0 \cap U_1} \\ & \uparrow & & & \downarrow \\ & s \otimes t & \dashrightarrow & & (\varphi_0^{-1})^\#(\varphi_0^\#(s)|_{U_0 \cap U_1} \varphi_1^\#(t)|_{U_0 \cap U_1}) \end{array}$$

So the composition  $\mathcal{O}_{k^1}(k^1) \otimes \mathcal{O}_{k^1}(k^1) \rightarrow \mathcal{O}_{k^1}(k^1 - \{0\})$  yields

$$s \otimes t \mapsto s \cdot t \circ \varphi_1 \circ \varphi_0^{-1}|_{k^1 - \{0\}} \quad (1.48)$$

At the level of polynomial rings (see lemma (1.2.7)), this map is

$$k[x] \otimes k[y] \rightarrow k[u]_u \cong k[u, u^{-1}] : f \otimes g \mapsto f(u)g(u^{-1}) \quad (1.49)$$

which is clearly surjective. **TODO:** finish the argument.

## 1.5 Lecture 6,7,8

**Remark 1.5.1.** Take the product topology on  $k^1 \times k^1$ , where  $k^1$  is equipped with Zariski topology. Let  $U = k^1 - S$  and  $V = k^1 - T$  ( $S, T$  finite) be two open sets. Then  $U \times V = k^1 \times k^1 - (S \times k^1 \cup k^1 \times T)$  and in general, for a family of open sets  $U_\lambda, V_\lambda$ , we have the equality.

$$\begin{aligned} \left(\bigcup_{\lambda} U_{\lambda} \times V_{\lambda}\right)^c &= \bigcap_{\lambda} S_{\lambda} \times k^1 \cup k^1 \times T_{\lambda} \\ &= \left(\bigcap_{\lambda} S_{\lambda} \times k^1\right) \cup \left(k^1 \times \bigcap_{\lambda} T_{\lambda}\right) \end{aligned} \quad (1.50)$$

where  $S_{\lambda}, T_{\lambda}$  are all finite. So any closed set  $F \subset k^1 \times k^1$  is of the form

$$F = S \times k^1 \cup k^1 \times T, \#S, \#T < \infty \quad (1.51)$$

in the product topology. In particular, if  $\#k = \infty$ , then  $V(X, Y) = \{0\}$  is not closed in the product topology.

**TODO:** remains to study lect 8, Nullstellensatz for projective spaces and “universal compactness” theorem.

### 1.5.1 Projective varieties

We define  $\mathbb{P}^m = (k^{m+1} \setminus \{0\}) / \sim$ . As usual, define charts  $\varphi_i : D_+(X_i) \rightarrow k^m : [x_0, \dots, x_m] \mapsto (\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_m}{x_i})$ . Then we see that the transition maps  $\varphi_i \varphi_j^{-1} : D(X_i) \subset k^m \rightarrow D(X_j) \subset k^m$  are isomorphisms of affine varieties. Furthermore,  $\mathbb{P}^m$  has a topology induced by the open affine covering  $\mathbb{P}^m = \bigcup_{i=0}^m D_+(X_i)$  where we set  $U \subset \mathbb{P}^m$  to be open if  $U \cap D_+(X_i) \subset D_+(X_i)$  is open  $\forall i$ . With this definition it's clear that  $D_+(X_i)$  is open, because  $\varphi_i(D_+(X_i X_j)) = D(X_j) \subset k^m$  is open  $\forall j$ . The sheaf of regular function is defined by using the charts:

$$\mathcal{O}_{\mathbb{P}^m}(U) = \{f : U \rightarrow k : f \varphi_i^{-1} \in \mathcal{O}_{k^m}(\varphi_i(U)), \forall i\} \quad (1.52)$$

In this condition, we see that  $\mathbb{P}^m$  is an abstract variety.

**Proposition 1.5.2.** Let  $\varphi_{ij} = \varphi_j \varphi_i^{-1} : \varphi_i(D(X_i X_j)) \rightarrow \varphi_j(D(X_i X_j))$ . Write  $U_{ij} = \varphi_i(D(X_i X_j)) \subset k^m$ . Then  $\varphi_{ij} : U_{ij} \rightarrow U_{ji}$  satisfies all the conditions of theorem (1.4.1). We claim that  $\mathbb{P}^m$  is isomorphic to the variety obtained by glueing all  $k^m$  through the isomorphisms  $\varphi_{ij}$ .



Proof: We have a well defined bijection

$$\mathbb{P}^m \rightarrow X = \coprod_{i=0}^m \{i\} \times k^m / \sim : D_+(X_i) \ni [x_0, \dots, x_m] \mapsto [\varphi_i([x_0, \dots, x_m])]_i \quad (1.53)$$

where we write  $(\underline{x})_i$  for an element of  $\{i\} \times k^m$ . It remains to show that this map is an isomorphism. However, at the level of charts we have the identity map.  $\square$

## 1.5.2 Projective Nullstellensatz

**Proposition 1.5.3.** Let  $I \subset k[X_0, \dots, X_m]$  be an ideal generated by homogeneous polynomials. Then  $F = \sum_i F_i \in I \iff F_i \in I \forall i$  where  $F_i$  are the homogeneous components of  $F$ . In each case we say that the ideal  $I$  is *homogeneous*.

**Proposition 1.5.4.** An ideal  $I$  is homogeneous if and only if  $\sqrt{I}$  is homogeneous.

**Notation 1.5.5.** Let  $I \subset k[X_0, \dots, X_m]$  be a homogeneous ideal, then we denote

$$V_p(I) = \{[x_0, \dots, x_m] \in \mathbb{P}^m : F(x_0, \dots, x_m) = 0, \forall F \in I \text{ homogenous}\}. \quad (1.54)$$

Furthermore, if  $V \subset \mathbb{P}^m$  is a subset, we write

$$I_p(V) = \langle S \rangle, \quad S = \{F \in k[X_0, \dots, X_m] : F \text{ is homogeneous and } F(p) = 0, \forall p \in V\}. \quad (1.55)$$

**Theorem 1.5.6.** Let  $I \subset k[X_0, \dots, X_m]$  be a homogeneous ideal. Then

- (1)  $V_p(I) = \emptyset \iff \sqrt{I} \supset (X_0, \dots, X_m)$
- (2) If  $V_p(I) \neq \emptyset$ , then  $I_p V_p(I) = \sqrt{I}$

Proof: For the first statement, we observe that  $V_p(I) = \emptyset \iff V_a(I) = \{0\}$ . However, by the Nullstellensatz we have

$$\sqrt{I} = I_a V_a(I) = I_a(\{0\}) = (X_0, \dots, X_m). \quad (1.56)$$

For the second statement, we observe that if  $V_p(I) \neq \emptyset$ , then

$$\begin{aligned} I_p V_p(I) &= \langle F \text{ homogeneous} : F[x_0, \dots, x_m] = 0, \forall [x_0, \dots, x_m] \in V_p(I) \rangle \\ &= \langle F \text{ homogeneous} : F[x_0, \dots, x_m] = 0, \forall (x_0, \dots, x_m) \in V_a(I) \setminus \{0\} \rangle \\ &= \langle F \text{ homogeneous} : F(x_0, \dots, x_m) = 0, \forall (x_0, \dots, x_m) \in V_a(I) \rangle \\ &= \langle F \text{ homogeneous} : F \in I_a V_a(I) \rangle \\ &= \langle F \text{ homogeneous} : F \in \sqrt{I} \rangle = \sqrt{I}. \end{aligned} \quad (1.57)$$

The last line follows because  $\sqrt{I}$  is a homogenous ideal.  $\square$

### 1.5.3 Segre embedding

**Remark 1.5.7.** Here, to ease notation, we will write  $x = [x_0, \dots, x_m]$  sometimes.

The Segre embedding is the map

$$s : \mathbb{P}^m \times \mathbb{P}^n \rightarrow \mathbb{P}^{(m+1)(n+1)-1}$$

$$([x_0, \dots, x_m], [y_0, \dots, y_n]) \mapsto \begin{bmatrix} x_0 y_0 & x_0 y_1 & \dots & x_0 y_n \\ x_1 y_0 & x_1 y_1 & \dots & x_1 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_m y_0 & x_m y_1 & \dots & x_m y_n \end{bmatrix} \quad (1.58)$$

where we parametrize  $\mathbb{P}^{(m+1)(n+1)-1}$  with projective coordinates  $z_{ij}$ ,  $0 \leq i \leq m, 0 \leq j \leq n$ . We claim that  $s(\mathbb{P}^m \times \mathbb{P}^n)$  is a closed subset. In fact, if we let  $z_{ij} = x_i y_j$ , then we have  $z_{ij} z_{pq} = z_{iq} z_{pj}$  and

$$s(\mathbb{P}^m \times \mathbb{P}^n) \subset V = V_p(z_{ij} z_{pq} - z_{pj} z_{iq}, \forall p, q, i, j) \quad (*). \quad (1.59)$$

Conversely, if we take  $z \in V$ , then  $z_{ij} \neq 0$  for some  $i, j$ . So we assume  $z_{ij} = 1$ . Then we see that  $z_{pq} = z_{pj} z_{iq}$ , so we can take  $x_p = z_{pj}$  and  $y_q = z_{iq}$  to conclude that  $s(x, y) = z$ . Then we conclude the equality of (\*). Furthermore, we claim that this map is injective. Let  $s(x, y) = s(x', y')$ , then  $x_p y_q = \lambda x'_p y'_q$ . Fix  $q$  such that  $y_q \neq 0$ , then  $x_p = (\frac{\lambda y'_q}{y_q}) x'_p$ ,  $\forall p$  and  $x = x'$ . In a similar way we see that  $y = y'$ .

As a corollary, we see that  $s$  maps  $\mathbb{P}^m \times \mathbb{P}^n$  bijectively to a closed subset  $V \subset \mathbb{P}^{(m+1)(n+1)-1}$ . We put a structure of abstract variety on  $\mathbb{P}^m \times \mathbb{P}^n$  in such a way that  $s$  is an isomorphism onto  $V$ . So we see for example, that

$$\mathcal{O}_{\mathbb{P}^m \times \mathbb{P}^n}(U) = (- \circ s)(\mathcal{O}_V(s(U))). \quad (1.60)$$

In a concrete way, a section  $s \in \mathcal{O}_{\mathbb{P}^m \times \mathbb{P}^n}$  is a quotient of bihomogeneous polynomials of bi degree  $(d, d)$  each one, for some  $d \geq 0$ . In fact, if  $F(z) \in k[\underline{z}]$  is homogeneous, then

$$F(z) = \sum a_\alpha \prod_{i,j} z_{ij}^{\alpha_{ij}}, \quad (i, j) \in \{0, \dots, m\} \times \{0, \dots, n\}. \quad (1.61)$$

Therefore

$$\begin{aligned} F \circ s(x, y) &= \sum a_\alpha \prod_{i,j} x_i^{\alpha_{ij}} y_j^{\alpha_{ij}} \\ &= \sum a_\alpha \left( \prod_i x_i^{\sum_j \alpha_{ij}} \right) \left( \prod_j y_j^{\sum_i \alpha_{ij}} \right) \end{aligned} \quad (1.62)$$

where  $\sum_i \sum_j \alpha_{ij} = \sum_j \sum_i \alpha_{ij} = |\alpha| = d$ , the degree of  $F$ . It's easy to see that a quotient of bihomogeneous polynomials of bidegree  $(d, e)$  each one, is also a section of  $\mathcal{O}_{\mathbb{P}^m \times \mathbb{P}^n}$ . In

fact, let  $G/H$  be a quotient of two bihomogeneous polynomials of bidegree  $(d, e)$  each one. Then, in  $D(x_i) \times D(y_j)$  we have

$$\frac{G}{H} = \frac{Gx_i^e y_j^d}{Hx_i^e y_j^d}. \quad (1.63)$$

So we can assume both  $G, H$  to be of the same bidegree  $(d + e, d + e)$ . Moreover, let  $F$  be a bihomogeneous polynomial of bidegree  $(d, d)$ , then

$$\begin{aligned} F(x, y) &= \sum a_{\alpha\beta} x^\alpha y^\beta \\ &= \sum a_{\alpha\beta} x_{i_1} y_{j_1} \cdots x_{i_d} y_{j_d} \quad (1 \leq i_k \leq m, 1 \leq j_k \leq n) \\ &= \sum a_{\alpha\beta} z_{i_1 j_1} \cdots z_{i_d j_d} \end{aligned} \quad (1.64)$$

is homogeneous of degree  $d$ .

**Lemma 1.5.8.** The variety  $\mathbb{P}^m \times \mathbb{P}^n$  constructed via the Segre embedding is a product.

Proof: Let  $f : Z \rightarrow \mathbb{P}^m$  and  $g : Z \rightarrow \mathbb{P}^n$  be morphisms. We draw the following diagram

$$\begin{array}{ccc} & D(x_i) \times D(y_j) & \xrightarrow{s} D(z_{ij}) \cap V \\ & \uparrow (f,g)|_U & \nearrow (\varphi_i, \psi_j) \\ U = f^{-1}(D(x_i)) \cap g^{-1}(D(y_j)) & & \\ & \downarrow (\varphi_i f|_U, \psi_j g|_U) & \nwarrow s \circ (\varphi_i^{-1}, \psi_j^{-1}) \\ & k^m \times k^n & \end{array}$$

We see that  $\varphi_i f|_U, \psi_j g|_U$  are morphisms, so  $(\varphi_i f|_U, \psi_j g|_U)$  is a morphism because it's a mapping into a product of affine varieties and we can apply the affine criteria. So, if we show that  $s \circ (\varphi_i^{-1}, \psi_j^{-1})$  is a morphism, we are done. However, we see that

$$\xi_{ij} \circ s \circ (\varphi_i^{-1}, \psi_j^{-1}) : k^m \times k^n \rightarrow k^{(m+1)(n+1)} : (x, y) \mapsto (\dots, x_i, \dots, y_j, \dots, x_i y_j, \dots) \quad (1.65)$$

which is clearly a morphism.  $\square$

**Remark 1.5.9.** The above proof also shows that  $(\varphi_i, \psi_j) : D(x_i) \times D(y_j) \rightarrow k^m \times k^n$  is a morphism. Furthermore, let  $X \subset \mathbb{P}^m, Y \subset \mathbb{P}^n$  be two closed subsets. Then, is  $X \cap D(x_i) \times Y \cap D(y_j) \subset D(x_i) \times D(y_j)$  closed? We have that  $\varphi_i(X \cap D(x_i)) \subset k^m, \psi_j(Y \cap D(y_j)) \subset k^n$  are closed. Therefore, the product

$$\varphi_i(X \cap D(x_i)) \times \psi_j(Y \cap D(y_j)) = (\varphi_i, \psi_j)(X \cap D(x_i) \times Y \cap D(y_j)) \quad (1.66)$$

is closed in  $k^m \times k^n$ . Here we use the standard observation that  $V(S) \times V(T) = V(S \cup T)$ , where  $S \subset k[\underline{X}], T \subset k[\underline{Y}]$  with both rings sitting inside  $k[\underline{x}, \underline{y}]$ . Since  $(\varphi_i, \psi_j)$  is a homeomorphism, this proves that the answer is positive.

**Corollary 1.5.10.** Let  $X \subset \mathbb{P}^m, Y \subset \mathbb{P}^n$  be two closed subsets. Then  $X \times Y \subset \mathbb{P}^m \times \mathbb{P}^n$  is closed.

**Remark 1.5.11.** The product of the affine spaces  $k^m$  and  $k^n$  is  $k^m \times k^n \cong k^{m+n}$ . So, if  $f : Z \rightarrow k^m, g : Z \rightarrow k^n$  are morphisms, then we draw the diagram.

$$\begin{array}{ccccc}
 & & k^m \times k^n & & \\
 & \nearrow (f,g) & \downarrow p_1 & \searrow p_2 & \\
 Z & & k^m & & k^n \\
 & \searrow f & \downarrow g & \nearrow & \\
 & & & & 
 \end{array}$$

However,  $f$  being a morphism means that  $x_i \circ f$  and  $y_j \circ g$  are morphisms for all  $i, j$  ( $1 \leq i \leq m, 1 \leq j \leq n$ ), where  $x_i$  and  $y_j$  are the standard coordinate functions. However, the set of coordinate functions of the product  $k^m \times k^n$  is  $(x_1, \dots, x_m, y_1, \dots, y_n)$  because of the identification  $k^m \times k^n \cong k^{m+n}$ . So, again we see that

$$(x_1 \circ f, \dots, x_m \circ f, y_1 \circ g, \dots, y_n \circ g) = (f, g) \quad (1.67)$$

is a morphism. Here we used lemma (1.3.11) several times.

#### 1.5.4 Grassmanians and Flag varieties

Let  $\text{Gr}_d(V)$  be the set of  $d$ -dimensional subspaces of  $V$ . We have the following embedding

$$\psi : \text{Gr}_d(V) \hookrightarrow \mathbb{P}(\bigwedge^d V) : W = \text{Span}(w_1, \dots, w_d) \mapsto [w_1 \wedge \dots \wedge w_d]. \quad (1.68)$$

The map is well defined, in fact, if  $\{w_i\}_{i=1}^d$  and  $\{w'_i\}_{i=1}^d$  are two basis of  $W$  and  $w'_i = \sum_j b_{ij} w_j$ , then  $w'_1 \wedge \dots \wedge w'_d = \det(b_{ij})_{i,j=1}^d w_1 \wedge \dots \wedge w_d$ . Now, let  $v_1, \dots, v_n$  be a basis of  $V$  and  $u_1, \dots, u_d \in V$  be arbitrary vectors. Write  $u_i = \sum_j a_{ij} v_j$ , then

$$\begin{aligned}
 u_1 \wedge \dots \wedge u_d &= \left( \sum_{j_1=1}^n a_{1j_1} v_{j_1} \right) \wedge \dots \wedge \left( \sum_{j_d=1}^n a_{dj_d} v_{j_d} \right) \\
 &= \sum_{j_1=1}^n \dots \sum_{j_d=1}^n a_{1j_1} \dots a_{dj_d} v_{j_1} \wedge \dots \wedge v_{j_d} \\
 &= \sum_{j_1 < \dots < j_d} \det(a_{ij_k})_{i,k=1}^d v_{j_1} \wedge \dots \wedge v_{j_d}
 \end{aligned} \quad (1.69)$$

So we see that  $u_1 \wedge \dots \wedge u_d = 0 \iff \text{Rank}(a_{ij}) < d \iff \{u_1, \dots, u_d\}$  is linearly dependent. So, if

$$w_1 \wedge \dots \wedge w_d = \lambda w'_1 \wedge \dots \wedge w'_d \neq 0, \quad (1.70)$$

then  $u \wedge w_1 \wedge \cdots \wedge w_d = 0 \iff u \wedge w'_1 \wedge \cdots \wedge w'_d = 0$ , i.e.,  $u \in \text{Span}\{w_i\}_{i=1}^d \iff u \in \text{Span}\{w'_i\}_{i=1}^d$ . Now, let  $\mathcal{J} = \{(j_1, \dots, j_d) : 1 \leq j_1 < \cdots < j_d \leq n\}$  and if  $J = (j_1, \dots, j_d) \in \mathcal{J}$ , write  $v_J = v_{j_1} \wedge \cdots \wedge v_{j_d}$ . Then we have the basis  $\{v_J\}_{J \in \mathcal{J}}$  of  $\Lambda^d V$  and we write  $X_J$  for the corresponding coordinate functions.

Let  $W \in \text{Gr}_d(V)$ , we see that  $\psi(W) \in D(X_{(1, \dots, d)})$  if and only if  $\det(a_{ij})_{i,j=1}^d \neq 0$ . In this case, by row reduction, we can assume that  $a_{ij} = \delta_{ij}$ ,  $(1 \leq i, j \leq d)$ , so  $W$  has a basis consisting of vectors of the form  $w_i = v_i + \sum_{j>d} a_{ij} v_j$ ,  $i = 1, \dots, d$ . Now define  $J_{pq}$  as the tuple  $(j_1, \dots, j_d)$  such that  $j_k = k$  ( $k < p$ ),  $j_k = k + 1$ , ( $p \leq k < d$ ) and  $j_d = q \in \{d + 1, \dots, n\}$ . With this notation, we see that

$$a_{pq} = \pm \det(a_{ij_k})_{j,k=1}^d = \pm X_{J_{pq}}(w_1 \wedge \cdots \wedge w_d). \quad (1.71)$$

So, for any  $J = (j_1, \dots, j_d) \in \mathcal{J}$ , the expression  $\det(a_{ij_k})_{i,k=1}^d$  is a polynomial  $f_J(X) \in k[X_{J_{pq}} : 1 \leq p \leq d, d < q \leq n] \subset k[X_J : J \in \mathcal{J} \setminus \{(1, \dots, d)\}]$  and we see that

$$\begin{aligned} \varphi_{(1, \dots, d)}(\text{Im}(\psi) \cap D(X_{(1, \dots, d)})) \\ = V(X_J - f_J(X) : \forall J = (j_1, \dots, j_d) \in \mathcal{J} \setminus \{(1, \dots, d), J_{pq} : p, q\}) \end{aligned} \quad (1.72)$$

which is closed, so  $\text{Im}(\psi) \cap D(X_{(1, \dots, d)})$  is closed. With the same argument applied to the other principal open subsets  $D(X_J)$ , we see that  $\text{Im}(\psi)$  is closed. It's important to note that the row echelon form of  $A = (a_{ij})$  gives an affine coordinate expression for  $\psi(W) \in U_{(1, \dots, d)}$ :

$$\varphi_{(1, \dots, d)}\psi(W) = (f_J(a))_{J \in \mathcal{J} \setminus \{(1, \dots, d)\}} \quad (1.73)$$

where  $f_J(a) = \det(a_{ij_k})_{i,k=1}^d$ ,  $J = (j_1, \dots, j_d)$ .

Now we define the Flag variety  $\mathcal{F}(V)$ , it's defined as the  $n$ -tuple of subspaces

$$(F_1, \dots, F_n) \in \text{Gr}_1(V) \times \cdots \times \text{Gr}_n(V) \quad (1.74)$$

such that  $F_i \subset F_{i+1}$ ,  $1 \leq i < n$ . It's a closed subset, so it's a projective variety in the above product. We prove this in two steps. The first one is to prove that

$$\{(F, F') : F \subset F'\} \subset \text{Gr}_d(V) \times \text{Gr}_{d+1}(V) \quad (1.75)$$

is a closed subset. As before, we simplify the problem by showing that the intersection

$$\{(F, F') : F \subset F'\} \cap U_{(1, \dots, d)} \times U_{(1, \dots, d, d+1)} \quad (*) \quad (1.76)$$

is closed. So let  $F$  and  $F'$  be represented by matrices  $(a_{ij}) \in \text{Mat}_{d,n}(k)$  and  $(b_{pq}) \in \text{Mat}_{d+1,n}(k)$  in row echelon form, i.e.,  $a_{ij} = \delta_{ij}$ ,  $j \leq d$  and  $b_{pq} = \delta_{pq}$ ,  $q \leq d + 1$ . In this condition  $F \subset F'$  if and only if the matrix

$$M = \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \cdots & \vdots \\ b_{d+1,1} & \cdots & b_{d+1,n} \\ a_{11} & \cdots & a_{1n} \\ \vdots & \cdots & \vdots \\ a_{d1} & \cdots & a_{dn} \end{pmatrix} \quad (1.77)$$

has rank  $d + 1$ . In particular, this means that any  $d + 2$  minor of  $M$  must vanish, which is a polynomial condition on the coefficients  $a_{ij}, b_{pq}$ . By equation (1.71), we see that this defines a polynomial condition in the affine coordinate system of  $\varphi_{(1,\dots,d)} \times \varphi_{(1,\dots,d+1)}$ , hence intersection  $(*)$  is closed.

Now, for the second step, we use corollary (1.5.10) to conclude that

$$S_d = \text{Gr}_1(V) \times \cdots \times \text{Gr}_{d-1}(V) \times \{(F, F') \in \text{Gr}_d(V) \times \text{Gr}_{d+1}(V) : F \subset F'\} \times \cdots \times \text{Gr}_{d+2}(V) \times \cdots \times \text{Gr}_n(V) \quad (1.78)$$

is closed. Therefore

$$\mathcal{F}(V) = \bigcap_{d=1}^{n-1} S_d \quad (1.79)$$

is closed, so it's a projective variety.

**Proposition 1.5.12.** We have  $\dim(\text{Gr}_d(V)) = d(n - d)$ ,  $n = \dim(V)$ .

Proof: From equation (1.72), we see that at  $U_{(1,\dots,d)} \subset \mathbb{P}(\wedge^d V)$ , the Grassmanian  $\text{Gr}_d(V)$  is defined by  $V(I)$ , where

$$I = (X_J - f_J(X) : \forall J = (j_1, \dots, j_d) \in \mathcal{J} \setminus \{(1, \dots, d), J_{pq} : p, q\}) \quad (1.80)$$

Put any LEX order on  $k[X_J : J \in \mathcal{J} \setminus \{(1, \dots, d)\}]$ , with the property  $X_J > X_{J_{pq}}$  whenever  $J \notin \{J_{pq} : (p, q) \in \{1, \dots, d\} \times \{d+1, \dots, n\}\}$ . Then, we see that  $\sqrt{I} = I$  and

$$\frac{k[X_J : J \neq (1, \dots, d)]}{I} \cong k[X_{J_{pq}} : (p, q) \in \{1, \dots, d\} \times \{d+1, \dots, n\}] \quad (1.81)$$

which has transcendence degree  $d(n - d)$ . □

There is a simple way to represent an element  $F \in \mathcal{F}(V)$ . Fix a basis  $\{e_1, \dots, e_n\}$  of  $V$  and observe that we can find a basis of  $V$  such that  $F_i = \langle v_1, \dots, v_i \rangle$ . Now, assume that  $F \in U = U_{(1)} \times \cdots \times U_{(1,\dots,d-1)} \times U_{(1,\dots,d)}$ . Then if we write  $v_i = \sum_j a_{ij} e_j$ , we see that  $A = (a_{ij})$  has nonsingular principal minors  $M_{ii}$ ,  $i = 1 \dots, n$ . We call  $A$  a representative matrix of  $F$ .

Since  $a_{11} \neq 0$ , we can assume  $v_1 = e_1 + \sum_{j>1} a_{1j} e_j$ . Now, observe that if we only do elementary operations of the form  $R_j \rightarrow R_j + kR_i$ , ( $i < j$ ) and  $R_j \rightarrow kR_j$ , ( $k \neq 0$ ) (where  $R_i = (a_{i1}, \dots, a_{in})$  is the  $i$ th row of  $A$ ) on  $A$ , we still have  $A$  as a representative of the flag  $F$ . Using the hypothesis that the principal minors are nonsingular, we see that we can assume  $a_{ii} = 1, \forall i$ , while  $a_{ij} = 0$ , ( $i > j$ ). That is  $v_i = e_i + \sum_{j>i} a_{ij} e_j$ .

**Proposition 1.5.13.** We have  $\dim \mathcal{F}(V) = \frac{1}{2}n(n - 1)$ .

Proof: The above discussion shows that  $\mathcal{F}(V) \cap U$  is defined by the graph of a morphism  $c : k^{\frac{1}{2}n(n-1)} \rightarrow k^N$  for some  $N$ . This proves the proposition. □

**Example 1.5.14.** Suppose that  $F \in \mathcal{F}(k^3)$  has matrix representation

$$A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}. \quad (1.82)$$

Then, on the chart  $U = U_{(1)} \times U_{(1,2)} \times U_{(1,2,3)}$  we have coordinates:

$$(\{(2) \mapsto a, (3) \mapsto b\}, \{(1,3) \mapsto c, (2,3) \mapsto ac - b\}, \{*\}) \quad (1.83)$$

Then we see that  $U \cap \mathcal{F}(k^3)$  is the graph of the morphism  $k^3 \rightarrow k : (a, b, c) \mapsto ac - b$ .

## 1.6 Lecture 9

### 1.6.1 Induced function spaces

### 1.6.2 General definition of product

## 1.7 Lecture 10

In this section we let  $X$  to be an abstract variety and  $Y \subset X$  to be an irreducible subset (a point is a case of interest). Define

$$\mathcal{O}_{X,Y} = \varinjlim_{Y \cap U \neq \emptyset} \mathcal{O}_X(U) \quad (1.84)$$

where the directed system is defined by  $(\{\mathcal{O}_X(U)\}_{U \in \mathcal{U}}, \{\rho_{UV}\}_{U,V \in \mathcal{U}})$  where  $\mathcal{U}$  is some system of generators for the topology of  $X$ . To avoid set theoretical issues we can take  $\mathcal{U}$  to be the countable set of basic open sets of  $X$ .

**Notation 1.7.1.** We will denote the elements of  $\mathcal{O}_{X,Y}$  by  $\langle U, f \rangle$ . So  $\langle U, f \rangle = \langle V, g \rangle$  if and only if there exists  $W \subset U \cap V$  such that  $f|_W = g|_W$ . In particular, since  $Y$  is taken to be irreducible, we have that  $W \cap Y \subset U \cap V \cap Y$  is dense, so  $f|_{U \cap V \cap Y} = g|_{U \cap V \cap Y}$  by continuity of  $f$  and  $g$ .

**Proposition 1.7.2.** The ring  $\mathcal{O}_{X,Y}$  is a  $k$ -algebra and is a local ring.

Proof: The inclusion

$$k \hookrightarrow \mathcal{O}_{X,Y} : c \mapsto \langle X, c \rangle \quad (1.85)$$

shows that  $\mathcal{O}_{X,Y}$  is a  $k$ -algebra. Now, define the set

$$\mathfrak{m}_{X,Y} = \{\langle U, f \rangle : f|_{U \cap Y} = 0\}. \quad (1.86)$$

The condition  $f|_{U \cap Y} = 0$  is we defined on the class  $\langle U, f \rangle$ . In fact, if  $V \subset U$  is open, then  $f|_{V \cap Y} = 0 \Rightarrow f|_{U \cap Y} = 0$ , because  $V \cap Y \subset U \cap Y$  is dense. Furthermore  $\mathcal{O}_{X,Y}$  is clearly seen to be an ideal. Finally, we observe that

$$\mathcal{O}_{X,Y} - \mathfrak{m}_{X,Y} \subset \mathcal{O}_{X,Y}^*. \quad (1.87)$$

In fact, if  $\langle U, f \rangle$  is such that  $f(U \cap Y) \neq \{0\}$ , then the set  $V = \{q \in U : f(q) \neq 0\}$  is a nonempty open set of  $U$ . Therefore

$$\langle U, f \rangle = \langle V, f|_V \rangle = \langle V, \frac{1}{f}|_V \rangle^{-1} \in \mathcal{O}_{X,Y}^*. \quad (1.88)$$

□

**Proposition 1.7.3.** Let  $X \subset k^m$  be an affine algebraic set and  $Y \subset X$  be an irreducible subset. Denote by  $P = I(Y)/I(X) \subset A = k[X]$ , then  $P$  is a prime ideal and we have an isomorphism

$$\psi : A_P \rightarrow \mathcal{O}_{X,Y} : \overline{F}/\overline{G} \mapsto \langle D_X(G), F/G \rangle. \quad (1.89)$$

Proof: First observe that  $G \in I(Y) \iff G|_Y = 0 \iff D_Y(G) = \emptyset$ . Therefore  $\overline{G} \notin P \iff Y \cap D(G) \neq \emptyset$ . Furthermore, we have  $\overline{F}_1/\overline{G}_1 = \overline{F}_2/\overline{G}_2$  in  $A_P$  if and only if there exists  $G \notin P$  such that

$$\begin{aligned} \overline{G}(\overline{F}_1\overline{G}_2 - \overline{F}_2\overline{G}_1) &= 0 \in k[X] \iff \\ G(F_1G_2 - F_2G_1)|_X &= 0 \in \mathcal{O}_X(X) \iff \\ (F_1G_2 - F_2G_1)|_{D_X(G)} &= 0 \in \mathcal{O}_X(D_X(G)) \iff \\ \langle D_X(G), F_1/G_1 \rangle &= \langle D_X(G), F_2/G_2 \rangle. \end{aligned} \quad (1.90)$$

Then we see that  $\psi$  is well defined and injective. It remains to prove that it's surjective. To see this, let  $\langle U, f \rangle \in \mathcal{O}_{X,Y}$ , and let  $q \in U \cap Y$ . Since  $f : U \rightarrow k$  is regular, there exists  $U' \ni q, U' \subset U$  and polynomials  $F, G \in k[X], G(q') \neq 0 \forall q' \in U'$  such that  $f|_{U'} = F/G|_{U'}$ . In particular,  $q \in D_X(G) \cap Y$  and  $\langle U, f \rangle = \langle D_X(G), F/G \rangle$ . To see that  $P$  is prime, we observe that it's the image of the prime ideal  $I(Y)$  under the surjective morphism  $f : k[X_1, \dots, X_m] \rightarrow k[X]$  and  $P \subset \ker(f)$ . □

**Example 1.7.4.** Let  $X = k^1$  and  $Y = \{0\}$ , we have  $A = k[t]$  and  $P = I(Y) = (t)$ . So we see that  $A_P = k[t]_P = \{F/G : G \notin (t)\}$ . For the other side, we have

$$\mathcal{O}_{X,Y} = \{\langle D(G), F/G \rangle : G(0) \neq 0\} \quad (1.91)$$

because  $D(G) \cap Y \neq \emptyset \iff G(0) \neq 0$ .



## Chapter 2

# Algebraic Groups

This is some notes that i did based on the first chapter of the book [1].

### 2.1 Basic definitions

Some important observations:

- If  $\mathbb{F}$  is a finite field, the map which takes a polynomial in  $\mathbb{F}[X_1, \dots, X_n]$  and send to a function  $\mathbb{F}^n \rightarrow \mathbb{F}$  is not an isomorphism. For example, if  $\mathbb{F}$  has  $q$  elements, then  $\prod_{q \in \mathbb{F}} (X - q) \in \mathbb{F}[X] \mapsto 0$ . However, if  $\mathbb{F}$  is infinity and  $a_n X^n + a_{n-1} X^{n-1} + \dots + a_0 = 0$ , then we can select distinc elements  $\alpha_0, \dots, \alpha_n$  and we conclude that

$$\begin{pmatrix} a_0 & \dots & a_n \end{pmatrix} \begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha_0 & \alpha_1 & \dots & \alpha_n \\ \vdots & \vdots & \dots & \vdots \\ \alpha_0^n & \alpha_1^n & \dots & \alpha_n^n \end{pmatrix} = 0. \quad (2.1)$$

However  $\det(\alpha_i^j)_{i,j=0}^n = \prod_{i < j} (\alpha_i - \alpha_j) \neq 0$  and we conclude that  $a_0, \dots, a_n = 0$ . Now, if  $f \in \mathbb{F}[X_1, \dots, X_n]$ , then we can write (assumin  $X_1$  appears in f):

$$f = g_d(X_2, \dots, X_n)X_1^d + \dots + g_1(X_2, \dots, X_n)X_1 + g_0(X_2, \dots, X_n). \quad (2.2)$$

So the equality  $f = 0$ , the previous step for the case  $\mathbb{F}[X]$  and an induction on the number of variables yields  $g_d, \dots, g_0 = 0$ . Then we see that if  $|\mathbb{F}| = \infty$ , we can identify the ring  $\mathbb{F}[X_1, \dots, X_n]$  with polynomial functions  $\mathbb{F}^n \rightarrow \mathbb{F}$ .

As a consequence of this story, we see that if  $|\mathbb{F}| = \infty$ , then

$$I(\mathbb{F}^n) = (0) \subset \mathbb{F}[X_1, \dots, X_n]. \quad (2.3)$$

- (Hilbert basis theorem). The Hilbert basis theorem states that if  $A$  is noetherian, then  $A[X]$  is noetherian. Since  $\mathbb{F}[X]$  is a PID (and in consequence noetherian), we conclude by an induction on the number of variables that  $\mathbb{F}[X_1, \dots, X_n]$  is noetherian.

**Remark 2.1.1.** A ring is said to be noetherian if it satisfy the ascending chain condition, or equivalently, if all of its ideals are finitely generated.

- (Algebraic sets). Let  $S \subset \mathbb{F}[X_1, \dots, X_n]$ , then  $V(S) = \{a \in \mathbb{F}^n : f(a) = 0\}$ . It's immediate that  $V(S) = V(\langle S \rangle)$ . The relation  $V(\bigcup_{i \in I} S_i) = \bigcap_{i \in I} V(S_i)$  allow us to define the Zariski topology on  $\mathbb{F}^n$ . Moreover, for any subset  $V \subset \mathbb{F}^n$ , we let  $I(V) = \{f \in \mathbb{F}[X_1, \dots, X_n] : f(a) = 0, \forall a \in V\}$ . Then we have

$$\begin{cases} VI(W) = \overline{W} \\ IV(\mathfrak{a}) = \sqrt{\mathfrak{a}} \end{cases} \quad (2.4)$$

We also have the important relation

$$V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b}). \quad (2.5)$$

In fact

$$V(\mathfrak{a}) \cup V(\mathfrak{b}) \underset{\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{a}, \mathfrak{b}}{\subset} V(\mathfrak{a} \cap \mathfrak{b}) \underset{\mathfrak{a}\mathfrak{b} \subset \mathfrak{a} \cap \mathfrak{b}}{\subset} V(\mathfrak{a}\mathfrak{b}) = V(\{fg : f \in \mathfrak{a}, g \in \mathfrak{b}\}) \subset V(\mathfrak{a}) \cup V(\mathfrak{b}). \quad (2.6)$$

- (Coordinate rings). Let  $\mathfrak{a} = I(V)$  where  $V$  is an algebraic set. We define the coordinate ring  $A[V] = \mathbb{F}[X_1, \dots, X_n]/\mathfrak{a}$ .
- (LEX order). We define an ordering on  $\mathbb{Z}_{\geq 0}^n$  by  $\alpha > \beta$  if for the smallest  $i$  such that  $\alpha_i \neq \beta_i$  we have  $\alpha_i > \beta_i$ . So, with a fixed LEX on  $\mathbb{F}[X_1, \dots, X_n]$ , we can define the leading term of a polynomial:  $\text{LT}(f) = a_\alpha X^\alpha$ , where  $\alpha$  is the smallest tuple, relative to LEX such that  $a_\alpha \neq 0$ . In the case  $\text{LT}(f) = a_\alpha X^\alpha$ , we also write  $\text{LM}(f) = X^\alpha$ .
- (Groebner Basis). Let  $\mathfrak{a}$  be an ideal in  $\mathbb{F}[X_1, \dots, X_n]$ . A groebner basis of  $\mathfrak{a}$  is a finite subset  $G \subset \mathfrak{a}$  such that  $(\text{LT}(g) : g \in G) = \text{LT}(\mathfrak{a})$ .

**Example 2.1.2.** Let  $R = \mathbb{F}[X_1, \dots, X_n, Y_1, \dots, Y_n]$  with LEX:  $X_1 > \dots > X_n > Y_1 > \dots > Y_n$  and define an ideal  $\mathfrak{a} = (X_1 - g_1, \dots, X_n - g_n)$  for  $g_i \in \mathbb{F}[Y_1, \dots, Y_n]$ . Since  $\text{LT}(X_i - g_i) = X_i$ , it's clear that  $(X_1, \dots, X_n) \subset \text{LT}(\mathfrak{a})$ . Now, if  $f \in R$  and  $X_j \nmid \text{LT}(f)$ ,  $\forall j$ , then  $f \in \mathbb{F}[Y_1, \dots, Y_n]$  because its smallest term has power  $\alpha \in \{0\} \times \mathbb{Z}^n$ . So, we need to show that  $I \cap \mathbb{F}[Y_1, \dots, Y_n] = (0)$ . This can be seen by proving that the kernel of  $\varphi : R \rightarrow \mathbb{F}[Y] : X_i \mapsto g_i, Y_i \mapsto Y_i$  is  $I$ . The only nontrivial inclusion is  $\text{Ker}(\varphi) \subset I$ . To see this use a division algorithm to write

$$f = h_1(X_1 - g_1) + \dots + h_n(X_n - g_n) + r \quad (2.7)$$

where  $r \neq 0$  implies that no term of  $r$  is divisible by  $\text{LT}(X_i - g_i) = X_i$ ,  $\forall i$ . By applying  $\varphi$  on both sides of this equation we have the desired inclusion. From this we see that  $G = (X_1 - g_1, \dots, X_n - g_n)$  is a groebner basis for  $I$ .

- (Hilbert Polynomial).

**Definition 2.1.3.** Let  $I \subset \mathbb{F}[X_1, \dots, X_n]$  be an ideal. We define the affine Hilbert function:

$${}^a\text{HF}_I : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0} : s \mapsto \dim_{\mathbb{F}} \left( \frac{\mathbb{F}[X_1, \dots, X_n]_{\leq s}}{I_{\leq s}} \right). \quad (2.8)$$

**Example 2.1.4.** For example, the number  ${}^a\text{HF}_{(0)}(s)$  is the number of  $n$ -tuples of nonnegative integers whose sum is  $\leq s$ . To compute this, write the generating function

$$\prod_{i=1}^n \frac{1}{1 - X_i} = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^n} X^\alpha. \quad (2.9)$$

Now put  $X_i = t$ ,  $\forall i$ . We see that

$$(1 - t)^{-n} = \sum_{s \geq 0} \binom{-n}{s} (-1)^s t^s = \sum_{s \geq 0} \binom{n+s}{s} t^s. \quad (2.10)$$

So  ${}^a\text{HF}_{(0)}(s) = \binom{n+s}{s} = \frac{1}{n!} s^n + \dots$ .

**Example 2.1.5.** Let  $f \in \mathbb{F}[X_1, \dots, X_n]$  be a degree  $d$  polynomial. Then

$${}^a\text{HF}_{(f)}(s) = \binom{s+n}{s} - \binom{s-d+n}{s-d} = \frac{d}{(n-1)!} s^{n-1} + \dots. \quad (2.11)$$

An important property of the affine Hilbert function is the following:

**Theorem 2.1.6.** Let  $I \subset \mathbb{F}[X_1, \dots, X_n]$  be an ideal, then

$${}^a\text{HF}_I(s) = {}^a\text{HF}_{\text{LT}(I)}(s), \quad s \geq 0. \quad (2.12)$$

Another main result is the existence of the Hilbert polynomial.

**Theorem 2.1.7.** There exist a polynomial  ${}^a\text{HP}_I(t) \in \mathbb{Q}[t]$  such that

$${}^a\text{HP}_I(s) = {}^a\text{HF}_I(s), \quad \forall s \geq 0. \quad (2.13)$$

Proof (sketch):

- Find a Groebner basis  $G$  of  $I$
- Let  $M \subset \mathbb{Z}_{\geq 0}^n$  be such that  $\{X^\beta\}_{\beta \in M} = \{\text{LM}(g) : g \in G\}$
- Write

$$C(I) = \bigcap_{\beta \in M} C(\beta), \quad C(\beta) = \{\alpha \mid X^\beta \nmid X^\alpha\} \quad (2.14)$$

as a union of  $C(J, \tau) = \{\alpha \mid \alpha_j = \tau(j) \forall j \in J\}$ . To do this use

$$C(\beta) = \bigcup_{i=1}^n \bigcup_{t_i=0}^{\beta_i-1} C(\{i\}, \tau : i \mapsto t_i). \quad (2.15)$$

- Use the formula  $|C(J, \tau)_{\leq s}| = \binom{d+s-|\tau|}{d}$ ,  $d = n - |J|$ ,  $|\tau| = \sum_{j \in J} \tau(j)$  and the inclusion-exclusion principle to compute

$${}^a\text{HP}_I(s) = |C(I)_{\leq s}| = \left| \bigcup_{(J, \tau)} C(J, \tau)_{\leq s} \right|. \quad (2.16)$$

**Example 2.1.8.** Let  $I = (X_1 - X_3^2, X_2 - X_3^3)$  and fix the LEX order with  $X_1 > X_2 > X_3$ . Then we calculate

- $G = \{X_1 - X_3^2, X_2 - X_3^3\}$
- $M = \{(1, 0, 0), (0, 1, 0)\}$
- We have

$$\begin{aligned} C(I) &= C(\{1\}, \tau : 1 \mapsto 0) \cap C(\{2\}, \tau : 2 \mapsto 0) \\ &= C(\{1, 2\}, \tau : 1 \mapsto 0, 2 \mapsto 0). \end{aligned} \quad (2.17)$$

So we conclude that  $|C(I)|_{\leq s} = \binom{1+s-0}{1} = s + 1$ .

- (Algebraic dimension). Here we define the dimension of an algebraic set.

**Definition 2.1.9.** Let  $V \subset \mathbb{F}^n$  be an algebraic set. We define its dimension as  $\dim(V) = \deg {}^a\text{HP}_{I(V)}(t)$ . It has the property that it's the biggest integer  $d \geq 1$  such that there exist indexes  $1 \leq i_1 < \dots < i_d \leq n$  such that  $I(V) \cap \mathbb{F}[X_{i_1}, \dots, X_{i_d}] = (0)$ .

**Example 2.1.10.** Let  $H_f = V(f)$ ,  $f$  irreducible, then

$$\begin{aligned} \dim(H_f) &= \deg {}^a\text{HP}_{I(H_f)}(t) \\ &= \deg {}^a\text{HP}_{(f)}(t) \\ &= \deg\left(\frac{\deg(f)}{(n-1)!}t^{n-1} + \dots\right) \\ &= n - 1. \end{aligned} \quad (2.18)$$

The dimension has the expected property: If  $W \subset V$  is closed, then  $\dim(W) \leq \dim(V)$ . In fact

$$\begin{aligned} I(V) \subset I(W) &\Rightarrow \\ I(V)_{\leq s} \subset I(W)_{\leq s} &\Rightarrow \\ {}^a\text{HF}_{I(W)}(s) \leq {}^a\text{HF}_{I(V)}(s), \forall s \geq 0 &\Rightarrow \\ \deg {}^a\text{HF}_{I(W)}(t) \leq \deg {}^a\text{HF}_{I(V)}(t). \end{aligned} \quad (2.19)$$

- (Transcendence degree). Let  $A$  be a field which is also a finitely generated  $\mathbb{F}$ -algebra. We define the transcendence degree of  $A$  over  $\mathbb{F}$  (and write  $\partial_{\mathbb{F}}(A)$ ) to be largest size of an algebraically independent subset of  $A$  over  $\mathbb{F}$ .

**Example 2.1.11.** We have  $\partial_{\mathbb{F}}(\mathbb{F}[X_1, \dots, X_n]) = n$ . Indeed, let  $\{f_1, \dots, f_r\}$  be polynomials and suppose that  $r > n$ . Write

$$\begin{aligned} W_s &= \{f^\beta : \beta \in \mathbb{Z}_{\geq 0}^r, |\beta| \leq s\}, \quad d = \max\{\deg(f_i) : i = 1, \dots, r\} \\ V_s &= \{X^\alpha : |\alpha| \leq s\} \end{aligned} \quad (2.20)$$

Then  $W_s \subset V_{ds}$  and

$$|W_s| = \binom{r+s}{s} \stackrel{s \gg 0}{\geq} \binom{n+sd}{sd} = \dim(V_{sd}) \geq \dim(W_s). \quad (2.21)$$

So, for  $s \gg 0$  there exist  $\{a_\beta\}_\beta \subset \mathbb{F}$  such that  $\sum_\beta a_\beta f^\beta = 0$  and  $\{f_1, \dots, f_r\}$  is algebraically dependent.

**Theorem 2.1.12.** Let  $V$  be an algebraic set and write  $I = I(V)$ , then

$$\dim(V) = \partial_{\mathbb{F}}(A[V]) = \partial_{\mathbb{F}}\left(\frac{\mathbb{F}[X_1, \dots, X_n]}{I}\right). \quad (2.22)$$

Proof: Let  $d = \dim(V)$ , so there exist indexes  $1 \leq i_1 < \dots < i_d \leq n$  such that  $I \cap \mathbb{F}[X_{i_1}, \dots, X_{i_d}] = (0)$ . This is equivalent to say that  $[X_{i_1}], \dots, [X_{i_d}] \in A[V]$  are algebraically independent over  $\mathbb{F}$ . Therefore  $d \leq \partial_{\mathbb{F}}(A[V])$ . Now, suppose that  $\varphi_i = [f_i] \in A[V], i = 1, \dots, m$  are algebraically independent over  $\mathbb{F}$  and put  $N = \max \deg(f_i)$ , then the map

$$\mathbb{F}[Y_1, \dots, Y_m]_{\leq s} \mapsto \frac{\mathbb{F}[X_1, \dots, X_n]_{\leq Ns}}{I_{\leq Ns}} : g \mapsto g(\varphi_1, \dots, \varphi_m) \quad (2.23)$$

is injective. Then, by taking dimensions we get

$$\binom{m+s}{s} \leq {}^a\text{HF}_I(s), \quad \forall s \geq 0. \quad (2.24)$$

We know that both expressions are polynomials in  $s$ , so by taking degrees we conclude that  $m \leq \dim(V) = d$  and  $\partial_{\mathbb{F}}(A[V]) \leq d$ .  $\square$

With small changes in the above proof, we can verify that in the case of  $A[V]$  being an integral domain, we have also  $\partial_{\mathbb{F}}(K) = \dim(V)$ ,  $K = \text{Frac}(A[V])$ .

**Example 2.1.13.** Let  $C = V(X_1 - X_3^2, X_1 - X_3^3)$ , then  $I(C) = (X_1 - X_3^2, X_2 - X_3^3)$  and

$$\frac{\mathbb{F}[X_1, X_2, X_3]}{I(C)} \cong \mathbb{F}[X_3] \quad (2.25)$$

so  $\partial_{\mathbb{F}}(A[C]) = 1$ .

**Example 2.1.14.** The purpose of this example is to show that if  $V \subset \mathbb{F}^n, |V| = \infty$  is a linear space, then  $\dim(V) = \dim_{\mathbb{F}}(V)$ .

- Let  $\text{Ann}(V) = \text{Span}(f_1, \dots, f_m)$ . Since  $V$  has finite dimension, we have  $v \in V \iff f(v) = 0 \forall f \in \text{Ann}(V)$  and  $V = V(\{f_1, \dots, f_m\})$ .
- Let  $f_i = \sum a_{ij} X_j$ , where we identify  $X_j$  with the  $j$ th coordinate functions in  $\mathbb{F}^n$ . Now, if  $A = (a_{ij})$  and apply reduction to put  $A$  in the form:

$$\begin{pmatrix} a_{1j_1} & * & 0 & * & \cdots & 0 & \cdots \\ & & a_{2j_2} & * & \cdots & 0 & \cdots \\ & & & & \ddots & \cdots & \vdots \\ & & & & & 0 & \cdots & a_{mj_m} & \cdots \end{pmatrix} \quad (2.26)$$

with  $a_{ij_i} = 1$ ,  $j_1 < j_2 < \dots$ . Now put LEX with:  $X_{j_1} > \dots > X_{j_m} > X_j$   $j \notin \{j_1, \dots, j_m\}$  and write  $f_i = X_{j_i} + h_i$  with  $h_i \in \mathbb{F}[X_j : j \notin \{j_1, \dots, j_m\}]$ . Then we see that  $A[V] \cong \mathbb{F}[X_j : j \notin \{j_1, \dots, j_m\}]$  and  $(f_1, \dots, f_m)$  is prime. Therefore  $I(V) = (f_1, \dots, f_m)$ . Finally

$$\dim(V) = \dim_{\mathbb{F}}(A[V]) = \dim_{\mathbb{F}} \mathbb{F}[X_j : j \notin \{j_1, \dots, j_m\}] = \dim_{\mathbb{F}} V. \quad (2.27)$$

## 2.2 Tangent Spaces

- (Regular maps). Let  $V \subset \mathbb{F}^m, W \subset \mathbb{F}^n$  be algebraic sets. Then a map  $f : V \rightarrow W$  is said to be regular if there exist polynomials such that  $f = (f_1, \dots, f_m)$ . Such a map is continuous in the Zariski topology.

**Remark 2.2.1.** We can identify  $A[V]$  with the set of regular functions  $V \rightarrow \mathbb{F}$ . In this context, we have a contravariant functor:

$$\varphi : V \rightarrow W \rightsquigarrow \varphi^* : A[W] \rightarrow A[V] : [g] \mapsto [g] \circ \varphi \quad (2.28)$$

**Example 2.2.2.** Let  $\varphi : \mathbb{F} \rightarrow C : x \mapsto (x^3, x^2, x)$  and write  $C = (X_1 - X_3^2, X_2 - X_3^3)$ . Remember the isomorphism  $A[C] \rightarrow \mathbb{F}[Y] : [X_1] \mapsto Y^2, [X_2] \mapsto Y^3, [X_3] \mapsto Y$ . So we have

$$\varphi^* : A[C] \rightarrow A[\mathbb{F}] : [X_3] \mapsto [X_3] \circ \varphi. \quad (2.29)$$

The first map is  $[X_3] : C \rightarrow \mathbb{F} : (x^3, x^2, x) \mapsto x$  and the last map is  $\mathbb{F} \rightarrow C : x \mapsto x$ . At the algebraic level, the map  $\varphi^*$  is just the isomorphism  $\mathbb{F}[Y] \cong \mathbb{F}[X]$ .

- (Tangent spaces). Let  $V$  be an algebraic set and  $k_p = \mathbb{F}$  be the  $A[V]$ -module with action  $[f]\alpha \equiv f(p)\alpha$ . Let  $I(V) = (f_1, \dots, f_m)$ . We define

$$\begin{aligned} T_p(V) &= \{v \in \mathbb{F}^n : d(f)_p(v) = 0 \forall f \in I(V)\} \\ &= \{v \in \mathbb{F}^n : d(f_i)_p(v) = 0 \forall i\} \\ &= V(\{d(f_i)_p : i = 1, \dots, m\}) \end{aligned} \quad (2.30)$$

**Remark 2.2.3.** Caution ! The polynomials of  $f_i$  here are not the generators of  $V$ , but of  $I(V)$ . For example, if  $V = V(f)$ , we can conclude that  $T_p(V) = V(df_p)$  only if  $f$  is irreducible, i.e., if  $I(V) = (f)$ .

Here  $df_p \equiv \sum_{i=1}^n D_i(f)(p)X_i \in \mathbb{F}[X_1, \dots, X_n]$ . We have an isomorphism

$$T_p(V) \rightarrow \text{Der}_{\mathbb{F}}(A[V], k_p) : v \mapsto D_v; D_v([f]) = \sum_{i=1}^n [D_i(f)] \cdot v_i \in \mathbb{F}. \quad (2.31)$$

with inverse  $D \mapsto (D([X_i]))_{i=1}^n$ . Of course  $D_v([f]) = df_p(v)$ .

**Example 2.2.4.** Let  $C = (f_1, f_2)$ ,  $f_1 = X_1 - X_3^2$ ,  $f_2 = X_2 - X_3^3$  and  $p = (x^3, x^2, x) \in C$ . We have

$$d(f_1)_p = 1 - 2xX_3, \quad d(f_2)_p = 1 - 3x^2X_3 \quad (2.32)$$

and  $T_p(C) = \mathbb{F}\langle(3x^2, 2x, 1)\rangle$ .

**Example 2.2.5.** The determinant is an irreducible polynomial. Indeed, let  $\det \in \mathbb{F}[X_{ij} : 1 \leq i, j \leq n]$ . Then, if we apply Laplace theorem on the first row, we get

$$\det = A_{11}X_{11} - A_{12}X_{12} + \dots + (-1)^n A_{1n}X_n \quad (2.33)$$

where  $A_{ij}$  denotes the determinant of the matrix obtained by excluding the  $i$ th row and  $j$ th column. By induction on the number of variables, we assume that  $A_{ij}$  are irreducible polynomials. Now, suppose that  $\det = fg$ , where  $\deg(f), \deg(g) > 0$ . Choose LEX with  $X_{ij} > X_{k\ell}$  if  $i < k$  or if  $i = k$  and  $j < \ell$ . Then

$$\text{LT}(A_{11})X_{11} = \text{LT}(\det) = \text{LT}(f)\text{LT}(g). \quad (2.34)$$

Assume that  $X_{11} | \text{LT}(f)$  (so  $\text{LT}(g) < X_{11}$ ) and write  $f = AX_{11} + B$ ,  $\text{LT}(B) < X_{11}$ . Then we must have  $AgX_{11} = A_{11}X_{11}$  and  $A_{11} = Ag$ . Since  $\deg(g) > 1$  and  $A_{11}$  is irreducible, we must have  $g = A_{11}$  and  $A = 1$ . Now we get

$$-A_{12}X_{12} + \dots + (-1)^n A_{1n}X_n = BA_{11} \quad (2.35)$$

and  $-\text{LT}(A_{12})X_{12} = \text{LT}(B)\text{LT}(A_{11})$ . Since  $X_{22} | \text{LT}(A_{11})$ , we are led to  $X_{22} | \text{LT}(A_{12})$  which is a contradiction because  $X_{22}$  doesn't appear in  $A_{12}$  by definition.

Now, since  $\det$  is an irreducible homogeneous polynomial, the difference  $\det - 1$  must be irreducible. Therefore  $I(SL_n \mathbb{F}) = (\det - 1)$  and we can compute the tangent space

$T_I(SL_n) = V(d_I(\det))$ . We have

$$\begin{aligned}
d_I \det &= \sum_{k,\ell} D_{k\ell}(\det)(I) X_{k\ell} \\
&= \sum_{k,\ell} \sum_{\sigma \in S_n} D_{k\ell} \prod_{i=1}^n X_{i\sigma(i)}(I) X_{k\ell} \\
&= \sum_{k,\ell} \sum_{\sigma \in S_n} \delta_{\ell,\sigma(k)} \prod_{i \neq k} \delta_{i,\sigma(i)} X_{k\ell} \\
&= \sum_{k,\ell} \delta_{k\ell} X_{k\ell} \quad (\delta_{\ell,\sigma(k)} \prod_{i \neq k} \delta_{i,\sigma(i)} \neq 0 \text{ only if } \sigma = 1) \\
&= \sum_k X_{kk}.
\end{aligned} \tag{2.36}$$

Therefore  $T_I SL_n(\mathbb{F}) = \mathfrak{sl}_n \mathbb{F}$ , the Lie algebra of traceless matrices.

- (Gauss lemma). We state the Gauss lemma.

**Lemma 2.2.6.** Let  $R$  be a UFD ring (think in  $\mathbb{F}[X]$ ), then  $R[X]$  is a UFD.

Proof: Let  $f$  and  $K = \text{Frac}(R)$ . Since  $K[X]$  is a UFD, we have  $f = FG$ ,  $F, G \in K[X]$  with  $F$  irreducible and  $\deg(G) < \deg(f)$ . Since  $R$  is a UFD, there exist  $r, s \in R$  such that  $rF, sG \in R[X]$ . Decompose  $rs = t_1 \cdots t_k$  into irreducible elements. Then if  $F' = rF, G' = sG$ , we have

$$[F'][G'] = [rs][f] = 0 \in R/(t_i)[X] \tag{2.37}$$

because  $f \in R[X]$ . So, since  $R/(t_i)[X]$  is a domain, we see that  $t_i | F'$  or  $t_i | G'$ . Then we see that there exist a partition  $\{1, \dots, k\} = I \cup J$  such that  $F' / \prod_{i \in I} t_i, G' / \prod_{j \in J} t_j \in R[X]$ . It's clear that  $F'$  still irreducible, then an induction on degree yields the result.  $\square$

**Corollary 2.2.7.** Let  $f, g \in R = \mathbb{F}[X_1, \dots, X_n]$  and assume  $f$  irreducible with  $f \nmid g$ , then there exist  $F, G \in R$  such that

$$Gf + Fg = h \in \mathbb{F}[X_1, \dots, X_{n-1}] \setminus 0. \tag{2.38}$$

Proof: Let  $K = \text{Frac}(\mathbb{F}[X_1, \dots, X_{n-1}])$ , then  $f, g \in K[X_n]$  and there exist  $A, B \in K[X_n]$  such that  $Af + Bg = 1$  (by Gauss lemma  $f$  still irreducible as an element of  $K[X_n]$ ). Then there exist  $h \in \mathbb{F}[X_1, \dots, X_{n-1}]$  such that  $hA, hB \in \mathbb{F}[X_1, \dots, X_{n-1}]$ . So by writing  $G = hA, F = hB$  we are done.  $\square$

**Notation 2.2.8.** Let  $M$  be a  $A$ -module (where  $A$  is a  $k$ -algebra), then we define

$$\text{Der}_k(A, M) = \{D \in \text{Hom}_k(A, M) \mid D(xy) = x \cdot D(y) + y \cdot D(x), \forall x, y \in A\}. \tag{2.39}$$

Observe that since  $M$  is a  $A$ -module, it's also a  $k$ -vector space with action given by  $\alpha \cdot m \equiv (\alpha 1_A) \cdot m, \alpha \in k$ .



**Notation 2.2.9.** Let  $A = \mathbb{F}[X_1, \dots, X_n]/I$ , where  $M$  is a  $A$ -module. We write

$$\mathcal{T}_{A,M} = \{(v_1, \dots, v_n) \in M^n : \sum_i [D_i(f)] \cdot v_i = 0, \forall f \in I\}. \quad (2.40)$$

We have the isomorphism  $\mathcal{T}_{A,M} \rightarrow \text{Der}_{\mathbb{F}}(A, M) : v \mapsto D_v$  with inverse

$$D \mapsto (D(\overline{X_i}))_{i=1}^n. \quad (2.41)$$

Now consider the following context, let  $K = \text{Frac}(A)$  as  $A$ -module ( $A$  acts by multiplication). Then we have an isomorphism

$$\text{Der}_{\mathbb{F}}(K, K) \rightarrow \text{Der}_{\mathbb{F}}(A, K) : D \mapsto D|_A. \quad (2.42)$$

In particular we also have the isomorphism  $\mathcal{T}_{A,K} \cong \text{Der}_{\mathbb{F}}(K, K)$ . We observe that this is also a  $K$ -vector space isomorphism, it's a bijection compatible with the  $K$ -action. From this isomorphism, we see that  $\mathcal{T}_{A,K}$  only depends on  $K$ , and the fact that  $\text{Frac}(A) = K$ .

**Theorem 2.2.10.** Let  $A = \mathbb{F}[X_1, \dots, X_n]/I$  be a domain and  $K = \text{Frac}(A)$ . Then

$$\partial_{\mathbb{F}}(A) = \dim_K \text{Der}_{\mathbb{F}}(K, K). \quad (2.43)$$

Proof: Let  $d = \partial_{\mathbb{F}}(A) = \partial_{\mathbb{F}}(K)$ , then there exist  $z_1, \dots, z_{d+1}$  such that

- $K = \mathbb{F}(z_1, \dots, z_{d+1})$
- $\{z_1, \dots, z_d\}$  is algebraically independent
- $z_d$  is separable over  $\mathbb{F}(z_1, \dots, z_d)$

Let  $R = \mathbb{F}[z_1, \dots, z_{d+1}] \subset K$  and  $f \neq 0$  irreducible such that  $f(z_1, \dots, z_{d+1}) = 0$ . Then by corollary (2.2.7), we see that  $\alpha : K[Y_1, \dots, Y_{d+1}] \rightarrow K : g \mapsto g(z_1, \dots, z_{d+1})$  has kernel  $(f)$ . Therefore

$$\begin{aligned} \mathcal{T}_{R,K} &= \{(v_1, \dots, v_{d+1}) \in K^{d+1} : \sum_i [D_i(f)] \cdot v_i = 0\} \\ &= \{(v_1, \dots, v_{d+1}) \in K^{d+1} : \sum_i D_i(f)(z_1, \dots, z_{d+1}) \cdot v_i = 0\}. \end{aligned} \quad (2.44)$$

We see that  $D_i(f)(z_1, \dots, z_{d+1}) = 0 \Rightarrow D_i(f) \in (f)$  and  $D_i(f) = 0$ . Since  $f \neq 0$  and  $|\mathbb{F}| = \infty$ , we must have  $D_i(f) \neq 0$  for some  $i$ , so  $D_i(f)(z_1, \dots, z_{d+1}) \neq 0$  for some  $i$ . Then we conclude that

$$\dim_K \text{Der}_{\mathbb{F}}(K, K) = \dim_K \mathcal{T}_{R,K} = d = \partial_{\mathbb{F}}(K) = \partial_{\mathbb{F}}(A). \quad (2.45)$$

□

**Corollary 2.2.11.** If  $A = \mathbb{F}[X_1, \dots, X_n]/(f_1, \dots, f_m)$ , then

$$\partial_{\mathbb{F}}(A) = \dim_K \mathcal{T}_{A,K} = n - \text{Rank}_K([D_i(f_j)]). \quad (2.46)$$

**Example 2.2.12.** Let  $C = V(f_1, f_2)$ ,  $f_1 = X_1 - X_3^2$ ,  $f_2 = X_2 - X_3^3$ . We have

$$A[C] = \frac{\mathbb{F}[X_1, X_2, X_3]}{(f_1, f_2)} \rightarrow \mathbb{F}[X_3] : [g] \mapsto g(X_3^3, X_3^2, X_3). \quad (2.47)$$

So  $A[C]$  is a domain and we can define  $K = \text{Frac}(A[C])$ . We compute

$$\text{Rank}_K([D_i(f_j)]) = \text{Rank}_K \begin{pmatrix} 1 & 0 & -2[X_3] \\ 0 & 1 & -3[X_3] \end{pmatrix} = 2. \quad (2.48)$$

Then we conclude  $\dim(V) = 3 - 2 = 1$ .

- (differential of a regular map). Here we define the notion of differential of a regular map.

**Definition 2.2.13.** Let  $\varphi : V \rightarrow W$  be a regular map between algebraic varieties. Also, let  $p \in V, q = \varphi(p) \in W$ . We define:

$$d\varphi_p : \text{Der}_{\mathbb{F}}(A[V], k_p) \rightarrow \text{Der}_{\mathbb{F}}(A[W], k_q) : D_v \mapsto D_v \circ \varphi^* \quad (2.49)$$

At the level of tangent spaces, we have:

$$\begin{array}{ccc} \text{Der}_{\mathbb{F}}(A[V], k_p) & \xrightarrow{d\varphi_p} & \text{Der}_{\mathbb{F}}(A[W], k_q) \\ \uparrow & & \downarrow \\ T_p(V) & \xrightarrow{\quad \quad} & T_q(W) \\ & \uparrow \text{red} & \downarrow \text{red} \\ & v & (D_v \circ \varphi^*([Y_i]))_{i=1}^m \end{array}$$

$D_v \xrightarrow{\text{red}} D_v \circ \varphi^*$   
 $v \xrightarrow{\text{red}} (D_v \circ \varphi^*([Y_i]))_{i=1}^m$

In more details, if we write  $\varphi = (f_1, \dots, f_m)$ , then  $\varphi^*([Y_i]) = [f_i]$  and we get

$$D_v([f_i]) = \sum_{j=1}^m [D_j(f_i)] \cdot v_j = \sum_{j=1}^m D_j(f_i)(p) v_j. \quad (2.50)$$

Therefore

$$d\varphi_p(v) = (D_j(f_i)(p))_{i,j} \cdot v = J_p(\varphi) \cdot v. \quad (2.51)$$

**Example 2.2.14.** Let  $G \subset \mathbb{F}^{n^2}$  be an algebraic group and  $\mu : G \times G \rightarrow G$  be the product of matrices. We have  $G \times G \subset \mathbb{F}^{n^2} \times \mathbb{F}^{n^2} \cong \mathbb{F}^{2n^2}$ . Now, identify the first  $n^2$  coordinate functions with  $X_{ij}$  and the second  $n^2$  coordinate functions with  $Y_{ij}$ . So we have

$$\mu_{ij} = \sum_{\ell} X_{i\ell} Y_{\ell j}. \quad (2.52)$$

Then we have

$$\frac{\partial \mu_{ij}}{\partial X_{rs}}(I) = \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) = \delta_{ir} \delta_{sj}. \quad (2.53)$$

Therefore

$$d_{(I,I)}\mu(A, B) = \left( \sum_{r,s} \frac{\partial \mu_{ij}}{\partial X_{rs}}(I) A_{rs} + \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) B_{rs} \right)_{ij} = (A_{ij} + B_{ij}) = A + B. \quad (2.54)$$

TODO:

1. Prove that  $T_1 G$  is a Lie algebra. Remember the star operator.
2. Compute the Lie algebra of  $SL_n$  using the fact that  $\det -1$  is an irreducible polynomial.

## Chapter 3

### List of exercises

**Exercise 1.** Let  $U = k^2 - \{0\}$ , show that the set of regular functions  $U \rightarrow k$  are polynomials, then conclude that  $U$  is not an affine variety.

**Solution.** Let  $\varphi \in \mathcal{O}_U(U)$  be a regular function, then  $\varphi|_{D_{k^2}(X)} = f/X^m$  and  $\varphi|_{D_{k^2}(Y)} = g/Y^n$  and on the intersection  $D_{k^2}(X) \cap D_{k^2}(Y) = D_{k^2}(XY)$ , we have

$$Y^n f - X^m g = 0 \quad (*). \quad (3.1)$$

Since there exist an infinity number of points  $p \in D_{k^2}(XY)$  (assuming  $k$  is algebraically closed), we also have the equality  $(*)$  in  $k[X, Y]$ . Now, let  $f = \sum a_{ij} X^i Y^j$  and  $g = \sum b_{ij} X^i Y^j$ . Then, equality  $(*)$  means

$$a_{i,j-n} = a_{i-m,j}, \quad \forall i \geq m, j \geq n. \quad (3.2)$$

Therefore  $f/X^m = g/Y^n \in k[X, Y]$  and  $\varphi$  is a polynomial, because  $U = D_{k^2}(X) \cup D_{k^2}(Y)$ . In conclusion, we proved that the inclusion  $i : U \hookrightarrow k^2$  induces a ring isomorphism

$$i^\#(k^2) : k[X, Y] \rightarrow \mathcal{O}_U(U) : f \mapsto f|_U (**). \quad (3.3)$$

Now assume  $U = V(S)$ ,  $S \subset k[X, Y]$ , then  $f \in S \Rightarrow f|_U = 0$  and  $f = 0$  because of  $(**)$ . So  $S = \{0\}$  and  $U = k^2$ , which is a contradiction.  $\square$

# Bibliography

- [1] Meinolf Geck. *An introduction to algebraic geometry and algebraic groups*. Oxford University Press, 2013.
- [2] James E Humphreys. *Linear algebraic groups*. Vol. 21. Springer New York, 1975.
- [3] David Mumford et al. *The red book of varieties and schemes*. Vol. 1358. Springer Berlin, 1988.